# Topology Assignment Solutions

Topology: MAT307/M405/MAT803

Problem Sets 1–7

Submitted by **B23**

November 9, 2026

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|---|---|
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| Karmukilan | PS2: Q9–11 |
| Srijit Bhadra | PS2: Q12–15 |
| Arunodaya | PS3: Q1–5 |
| Pula Karan | PS3: Q6–10 |
| Dhrubojoti | PS4: Q1–4 |
| Aaditya | PS4: Q5–8 |
| Mayuri | PS5: Q1–4 |
| Prabhu | PS5: Q5–6 |
| Yashpal Prasad | PS6: Q1–3 |
| Indrajeet Kumawat | PS6: Q4–6 |
| Himanshu Sharma | PS6: Q7–9 |
| Karan Gangadharan | PS7: Q1–4 |

## Coverage

Solved problems are shown in **green**; pending ones in grey.

| Set | Problems | Solved |
|---|---|---|
| PS1 | **1 2 3 4 5 6 7 8 9 10** 11 12 13 14 **15 16 17** | 13/17 |
| PS2 | **1 2 3 4 5 6 7 8 9 10 11 12 13 14 15** | 15/15 |
| PS3 | **1 2 3 4 5 6 7 8 9 10** | 10/10 |
| PS4 | **1 2 3 4 5 6 7 8 9 10 11** | 11/11 |
| PS5 | **1 2 3 4 5 6** 7 8 9 10 | 6/10 |
| PS6 | **1 2 3 4 5 6 7 8 9** | 9/9 |
| PS7 | **1 2 3 4** 5 6 | 4/6 |
| | **Total solved** | **68/78** |

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# Problem Set 1

## Problem 1.1

Let $A$ be a set and $\mathcal{P}(A)$ be its power set. Prove that there is no injective map $f\colon \mathcal{P}(A) \to A$ and no surjective map $g\colon A \to \mathcal{P}(A)$.

**Solution** *submitted by* **Ansh**

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*Proof.* We establish both claims rigorously using Cantor's diagonalization argument, and we also show how the non-existence of an injection follows directly from the non-existence of a surjection.

**Part I: Non-existence of a surjective map $g\colon A \to \mathcal{P}(A)$.** Suppose, for the sake of contradiction, that there exists a surjective function $g\colon A \to \mathcal{P}(A)$. Define the diagonal subset $B \subseteq A$ by

$$B := \{x \in A \mid x \notin g(x)\}.$$

Note that this construction is well-defined for every set $A$ (including $A = \emptyset$, in which case $\mathcal{P}(\emptyset) = \{\emptyset\} \neq \emptyset$, so no map $g\colon \emptyset \to \mathcal{P}(\emptyset)$ exists at all).

Since $B \subseteq A$, we have $B \in \mathcal{P}(A)$. Because $g$ is assumed to be surjective, there exists at least one element $x_0 \in A$ such that

$$g(x_0) = B.$$

By the law of excluded middle, either $x_0 \in B$ or $x_0 \notin B$. We examine both cases:

(a) *Case 1 ($x_0 \in B$):* By the defining property of $B$, an element $x \in A$ belongs to $B$ if and only if $x \notin g(x)$. Consequently, $x_0 \in B \implies x_0 \notin g(x_0)$. However, since $g(x_0) = B$, this implies $x_0 \notin B$, contradicting our assumption that $x_0 \in B$.

(b) *Case 2 ($x_0 \notin B$):* Since $g(x_0) = B$, the assumption $x_0 \notin B$ means $x_0 \notin g(x_0)$. By the definition of $B$, any element of $A$ satisfying $x_0 \notin g(x_0)$ must belong to $B$; hence $x_0 \in B$, contradicting our assumption that $x_0 \notin B$.

In both cases we arrive at the logical contradiction $x_0 \in B \iff x_0 \notin B$. Therefore, our initial assumption is false, and no surjective map $g\colon A \to \mathcal{P}(A)$ exists.

**Part II: Non-existence of an injective map $f\colon \mathcal{P}(A) \to A$.** We provide two independent proofs of this result.

**Method 1 (Direct Diagonal Proof).** Suppose, for the sake of contradiction, that there exists an injective map $f\colon \mathcal{P}(A) \to A$. Consider the subset $C \subseteq A$ defined by

$$C := \{f(S) \in A \mid S \in \mathcal{P}(A) \text{ and } f(S) \notin S\}.$$

Since $C \subseteq A$, we have $C \in \mathcal{P}(A)$. Let $c := f(C) \in A$. We ask whether $c \in C$ or $c \notin C$:

(a) *Case 1 ($c \in C$):* By definition of $C$, there exists some subset $S \in \mathcal{P}(A)$ such that $c = f(S)$ and $f(S) \notin S$. Since $f(C) = c = f(S)$ and $f$ is injective, it follows that $C = S$. Substituting $S = C$ into $f(S) \notin S$ yields $c = f(C) \notin C$, a contradiction.

(b) *Case 2 ($c \notin C$):* Since $c = f(C)$, the statement $c \notin C$ is equivalent to $f(C) \notin C$. Taking $S = C \in \mathcal{P}(A)$ in the defining condition for $C$, we see that $f(C)$ satisfies the membership criterion for $C$, which implies $c = f(C) \in C$, a contradiction.

Thus, in both cases we obtain a contradiction. Hence, no injective map $f\colon \mathcal{P}(A) \to A$ can exist.

**Method 2 (Reduction to Part I).** Suppose $f\colon \mathcal{P}(A) \to A$ is injective. Since $\mathcal{P}(A)$ is always non-empty (as $\emptyset \in \mathcal{P}(A)$), its image $f(\mathcal{P}(A)) \subseteq A$ is non-empty, which implies $A \neq \emptyset$. Define a map $g\colon A \to \mathcal{P}(A)$ by

$$g(a) := \begin{cases} S & \text{if } a = f(S) \text{ for some } S \in \mathcal{P}(A), \\ \emptyset & \text{if } a \notin f(\mathcal{P}(A)). \end{cases}$$

Because $f$ is injective, for each $a \in f(\mathcal{P}(A))$ the preimage $S \in \mathcal{P}(A)$ with $f(S) = a$ is unique; thus $g$ is a well-defined function. Moreover, for every $S \in \mathcal{P}(A)$, we have $g(f(S)) = S$, showing that $g\colon A \to \mathcal{P}(A)$ is surjective. This directly contradicts Part I. Therefore, no injective map $f\colon \mathcal{P}(A) \to A$ exists. □

## Problem 1.2

Let $X$ be an infinite set. Is the collection $\mathcal{T}_\infty := \{U \subseteq X \mid X \setminus U \text{ is infinite or } U = \emptyset\}$ a topology on $X$? Justify your answer.

### Solution

*submitted by* **Ansh**

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**Answer.** No, $\tau_\infty$ is not a topology on $X$. It violates both the first axiom ($X \in \tau_\infty$) and the second axiom (closure under arbitrary unions; in fact, it even fails closure under finite unions).

**Justification.** For $\tau_\infty$ to be a topology on $X$, it must satisfy the three standard topological axioms:

(T1) $\emptyset \in \tau_\infty$ and $X \in \tau_\infty$;

(T2) $\tau_\infty$ is closed under arbitrary unions;

(T3) $\tau_\infty$ is closed under finite intersections.

We examine each axiom in turn:

**1. Violation of Axiom (T1) ($X \notin \tau_\infty$).** By explicit definition, $\emptyset \in \tau_\infty$. However, consider the whole space $U = X$. Since $X$ is an infinite set, $X \neq \emptyset$. Furthermore, the complement of $X$ in $X$ is

$$X \setminus X = \emptyset,$$

which has cardinality 0 and is therefore a finite set, not an infinite set. Because $U = X$ neither equals $\emptyset$ nor has an infinite complement in $X$, we have

$$X \notin \tau_\infty.$$

Thus, Axiom (T1) is violated.

**2. Violation of Axiom (T2) (Failure of closure under unions).** Even if one were to adjoin $X$ to the collection, $\tau_\infty$ fails to be closed under arbitrary unions—and even fails under the union of two sets:

- **Counterexample via arbitrary unions of singletons:** For any element $x \in X$, consider the singleton $U_x = \{x\}$. Since $X$ is infinite, removing a single element leaves an infinite set, so $X \setminus \{x\}$ is infinite and $\{x\} \in \tau_\infty$ for every $x \in X$. Fix a point $x_0 \in X$ and consider the union

$$U := \bigcup_{x \in X \setminus \{x_0\}} \{x\} = X \setminus \{x_0\}.$$

Since $X$ is infinite, $U = X \setminus \{x_0\}$ is non-empty (and $U \neq X$). Yet its complement is $X \setminus U = \{x_0\}$, which is a finite set of cardinality 1. Therefore, $U \notin \tau_\infty$.

- **Counterexample via a finite union of two sets:** Since $X$ is infinite, $X$ contains a countably infinite subset $C = \{x_1, x_2, x_3, \dots\}$ with $x_i \neq x_j$ for $i \neq j$. Partition $C$ into the set of even-indexed elements $E = \{x_2, x_4, x_6, \dots\}$ and odd-indexed elements $O = \{x_1, x_3, x_5, \dots\}$, both of which are infinite. Define

$$U_1 := X \setminus (E \cup \{x_1\}) \quad \text{and} \quad U_2 := X \setminus O.$$

Then $X \setminus U_1 = E \cup \{x_1\}$ is infinite, so $U_1 \in \tau_\infty$; similarly, $X \setminus U_2 = O$ is infinite, so $U_2 \in \tau_\infty$. However, by De Morgan's laws,

$$U_1 \cup U_2 = X \setminus \big((E \cup \{x_1\}) \cap O\big) = X \setminus \{x_1\}.$$

Since $U_1 \cup U_2 = X \setminus \{x_1\} \neq \emptyset$ and its complement $X \setminus (U_1 \cup U_2) = \{x_1\}$ is finite (of size 1), we have $U_1 \cup U_2 \notin \tau_\infty$. (Moreover, taking $V_1 = X \setminus E \in \tau_\infty$ and $V_2 = X \setminus O \in \tau_\infty$ gives $V_1 \cup V_2 = X \notin \tau_\infty$.)

**3. Remark on Axiom (T3) (Closure under finite intersections holds).** For completeness, note that if $U_1, U_2 \in \tau_\infty$, then either one of them is $\emptyset$ (so $U_1 \cap U_2 = \emptyset \in \tau_\infty$) or both $X \setminus U_1$ and $X \setminus U_2$ are infinite, in which case $X \setminus (U_1 \cap U_2) = (X \setminus U_1) \cup (X \setminus U_2)$ contains $X \setminus U_1$ and is therefore infinite, so $U_1 \cap U_2 \in \tau_\infty$. Nevertheless, because Axioms (T1) and (T2) fail, $\tau_\infty$ is not a topology on $X$.

## Problem 1.3

Let $X = \{a, b, c\}$. Determine which of the following collections of subsets form a valid topology on $X$. For those that fail, explicitly state which axiom or axioms are violated:

(i) $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$;

(ii) $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a, b\}, X\}$;

(iii) $\mathcal{T}_3 = \{\emptyset, \{a, b\}, \{b, c\}, X\}$.

### Solution

*submitted by* **Ansh**

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Recall that a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ is a topology on $X$ if and only if: (T1) $\emptyset, X \in \mathcal{T}$; (T2) $\mathcal{T}$ is closed under arbitrary unions; and (T3) $\mathcal{T}$ is closed under finite intersections.

**Part (i): $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$.**

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_1$ and $X \in \mathcal{T}_1$.
- **Axiom (T2) (Arbitrary Unions): Violated.** Taking the union of $\{a\} \in \mathcal{T}_1$ and $\{b\} \in \mathcal{T}_1$ yields

$$\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}_1.$$

- **Axiom (T3) (Finite Intersections):** Satisfied, since $\{a\} \cap \{b\} = \emptyset \in \mathcal{T}_1$, and the intersection of any element of $\mathcal{T}_1$ with $\emptyset$, $X$, or itself remains in $\mathcal{T}_1$.
- **Conclusion:** $\mathcal{T}_1$ is **not a valid topology** on $X$ because it violates the **closure under unions axiom (T2)**.

**Part (ii): $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a,b\}, X\}$.** Observe that the sets in $\mathcal{T}_2$ form a strictly nested chain under set inclusion:

$$\emptyset \subsetneq \{a\} \subsetneq \{a,b\} \subsetneq \{a,b,c\} = X.$$

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_2$ and $X \in \mathcal{T}_2$.
- **Axiom (T2) (Arbitrary Unions):** Satisfied. Because $\mathcal{T}_2$ is finite and totally ordered by inclusion ($U \subseteq V$ or $V \subseteq U$ for all $U, V \in \mathcal{T}_2$), the union of any non-empty subcollection of $\mathcal{T}_2$ equals the largest element of that subcollection, which belongs to $\mathcal{T}_2$ (and the empty union is $\emptyset \in \mathcal{T}_2$). Explicitly, the only non-trivial pairwise union is $\{a\} \cup \{a,b\} = \{a,b\} \in \mathcal{T}_2$.
- **Axiom (T3) (Finite Intersections):** Satisfied. For any non-empty finite subcollection of $\mathcal{T}_2$, its intersection equals the smallest element in that subcollection, which belongs to $\mathcal{T}_2$. Explicitly, $\{a\} \cap \{a,b\} = \{a\} \in \mathcal{T}_2$.
- **Conclusion: $\mathcal{T}_2$ is a valid topology** on $X$.

**Part (iii): $\mathcal{T}_3 = \{\emptyset, \{a,b\}, \{b,c\}, X\}$.**

- **Axiom (T1):** Satisfied, since $\emptyset \in \mathcal{T}_3$ and $X \in \mathcal{T}_3$.
- **Axiom (T2) (Arbitrary Unions):** Satisfied, because $\{a,b\} \cup \{b,c\} = \{a,b,c\} = X \in \mathcal{T}_3$, and any union involving $\emptyset$, $X$, or a single set remains in $\mathcal{T}_3$.
- **Axiom (T3) (Finite Intersections): Violated.** Taking the intersection of $\{a,b\} \in \mathcal{T}_3$ and $\{b,c\} \in \mathcal{T}_3$ yields

$$\{a,b\} \cap \{b,c\} = \{b\} \notin \mathcal{T}_3.$$

- **Conclusion:** $\mathcal{T}_3$ is **not a valid topology** on $X$ because it violates the **closure under finite intersections axiom (T3)**.

| Collection | (T1) | (T2) Unions | (T3) Intersections | Topology? |
|---|---|---|---|---|
| (i) $\mathcal{T}_1$ | Holds | Fails ($\{a\} \cup \{b\} = \{a,b\} \notin \mathcal{T}_1$) | Holds | **No** |
| (ii) $\mathcal{T}_2$ | Holds | Holds | Holds | **Yes** |
| (iii) $\mathcal{T}_3$ | Holds | Holds | Fails ($\{a,b\} \cap \{b,c\} = \{b\} \notin \mathcal{T}_3$) | **No** |

## Problem 1.4

Let $X$ be a topological space and $A \subset X$. Prove that $A$ is open in $X$ if and only if for each $x \in A$, $\exists$ an open set $U$ in $X$ such that $x \in U \subseteq A$.

### Solution

*submitted by* **Ansh**

*Proof.* Let $(X, \tau)$ be a topological space and let $A \subseteq X$. We prove both directions of the biconditional statement.

**Forward Direction ($\Longrightarrow$).** Suppose that $A$ is open in $X$, so $A \in \tau$. Let $x \in A$ be arbitrary. Choose $U := A$. Since $A$ is open in $X$, $U$ is an open set in $X$, and by reflexivity of set inclusion ($A \subseteq A$), we immediately have

$$x \in U \subseteq A.$$

Thus, for each $x \in A$, there exists an open set $U$ in $X$ such that $x \in U \subseteq A$.

**Reverse Direction ($\Longleftarrow$).** Conversely, suppose that for every $x \in A$, there exists an open set $U_x \in \tau$ such that $x \in U_x \subseteq A$. We claim that

$$A = \bigcup_{x \in A} U_x.$$

We establish this set equality by proving mutual inclusion:

1. *Proof that* $A \subseteq \bigcup_{x \in A} U_x$*:* Let $y \in A$ be arbitrary. By hypothesis, the corresponding open set $U_y$ satisfies $y \in U_y$. Since $U_y$ is one of the sets in the indexed family $\{U_x\}_{x \in A}$, it follows from the definition of the union that $y \in \bigcup_{x \in A} U_x$. Hence, $A \subseteq \bigcup_{x \in A} U_x$. (Note that if $A = \emptyset$, the empty union is $\emptyset$, so the inclusion holds vacuously.)
2. *Proof that* $\bigcup_{x \in A} U_x \subseteq A$*:* Let $y \in \bigcup_{x \in A} U_x$ be arbitrary. By definition of the union, there exists some index $x_0 \in A$ such that $y \in U_{x_0}$. By hypothesis, $U_{x_0} \subseteq A$, which implies $y \in A$. Hence, $\bigcup_{x \in A} U_x \subseteq A$.

Combining both inclusions yields $A = \bigcup_{x \in A} U_x$. Because each $U_x$ is an open set in $X$ and a topology $\tau$ is closed under arbitrary unions, we conclude that $A = \bigcup_{x \in A} U_x \in \tau$; that is, $A$ is open in $X$. □

## Problem 1.5

Let $U$ be an open set and $V$ be a closed set in $X$. Prove that $U \setminus V$ is open in $X$ and $V \setminus U$ is closed in $X$.

### Solution

*submitted by* **Ansh**

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*Proof.* Let $(X, \tau)$ be a topological space, let $U \subseteq X$ be an open set in $X$, and let $V \subseteq X$ be a closed set in $X$.

First, recall that for any two subsets $A, B \subseteq X$, the set difference $A \setminus B$ is given by the identity

$$A \setminus B = \{x \in X \mid x \in A \text{ and } x \notin B\} = A \cap (X \setminus B).$$

Furthermore, by the definition of a closed set in a topological space, a subset $F \subseteq X$ is closed in $X$ if and only if its complement $X \setminus F$ is open in $X$.

**Part I: Proof that $U \setminus V$ is open in $X$.** Applying the set-difference identity to $U \setminus V$, we have

$$U \setminus V = U \cap (X \setminus V).$$

Since $V$ is closed in $X$, its complement $X \setminus V$ is open in $X$ by definition. By hypothesis, $U$ is also open in $X$. By the finite intersection axiom of a topology, the intersection of two open sets in $X$ is open in $X$. Therefore, $U \setminus V = U \cap (X \setminus V)$ is open in $X$.

**Part II: Proof that $V \setminus U$ is closed in $X$.** We provide both a direct proof from the open-set axioms via De Morgan's laws and a dual proof via closed-set intersections:

**Method 1 (Direct proof via De Morgan's Laws).** To show that $V \setminus U$ is closed in $X$, it suffices by definition to prove that its complement $X \setminus (V \setminus U)$ is open in $X$. Using $V \setminus U = V \cap (X \setminus U)$

and De Morgan's laws, we compute:

$$\begin{aligned} X \setminus (V \setminus U) &= X \setminus \big(V \cap (X \setminus U)\big) \\ &= (X \setminus V) \cup \big(X \setminus (X \setminus U)\big) \\ &= (X \setminus V) \cup U. \end{aligned}$$

Since $V$ is closed in $X$, its complement $X \setminus V$ is open in $X$. By hypothesis, $U$ is also open in $X$. Because a topology is closed under unions, $(X \setminus V) \cup U$ is open in $X$. Consequently, $X \setminus (V \setminus U)$ is open in $X$, which by definition means that $V \setminus U$ is closed in $X$.

**Method 2 (Via intersections of closed sets).** Using the set-difference identity, we write

$$V \setminus U = V \cap (X \setminus U).$$

Since $U$ is open in $X$ and $X \setminus (X \setminus U) = U$, the set $X \setminus U$ is closed in $X$. We are also given that $V$ is closed in $X$. Since the intersection of any two closed sets in $X$ is closed in $X$, it follows immediately that $V \setminus U = V \cap (X \setminus U)$ is closed in $X$.

This completes the proof of both statements. □

## Problem 1.6

Let $X = \mathbb{N}$ and define $\mathcal{T}_1, \mathcal{T}_2 \subseteq \mathcal{P}(\mathbb{N})$ as follows:

(i) $\mathcal{T}_1 := \{\emptyset, \mathbb{N}, \{1\}, \{1,2\}, \{1,2,3\}, \{1,2,3,4\}, \dots\}$.

(ii) $\mathcal{T}_2 := \{\emptyset, \mathbb{N}, \mathbb{N} \setminus \{1\}, \mathbb{N} \setminus \{1,2\}, \mathbb{N} \setminus \{1,2,3\}, \mathbb{N} \setminus \{1,2,3,4\}, \dots\}$.

Determine whether $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

### Solution

*submitted by* **Sahil**

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Throughout, a topology on a set $X$ is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ such that (1) $\varnothing, X \in \mathcal{T}$, (2) arbitrary unions of members of $\mathcal{T}$ belong to $\mathcal{T}$, and (3) finite intersections of members of $\mathcal{T}$ belong to $\mathcal{T}$.

Let

$$A_n = \{1, 2, \dots, n\}, \qquad B_n = \mathbb{N} \setminus A_n \quad (n \geq 1).$$

Then the two collections in the question can be written as

$$\mathcal{T}_1 = \{\varnothing, \mathbb{N}\} \cup \{A_n : n \geq 1\}, \qquad \mathcal{T}_2 = \{\varnothing, \mathbb{N}\} \cup \{B_n : n \geq 1\}.$$

We verify the topology axioms separately.

**(a) The collection $\mathcal{T}_1$ is a topology.** Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_1$. For finite intersections, observe that

$$A_m \cap A_n = A_{\min\{m,n\}}.$$

Intersections with $\varnothing$ or $\mathbb{N}$ also belong to $\mathcal{T}_1$. Thus $\mathcal{T}_1$ is closed under finite intersections.

Now let $\{U_i\}_{i \in I}$ be any family of members of $\mathcal{T}_1$. If every $U_i$ is empty, then $\bigcup_{i \in I} U_i = \varnothing$. If at least one $U_i = \mathbb{N}$, the union is $\mathbb{N}$. In the remaining case, the nonempty sets in the family are of the form $A_n$. If their indices are unbounded in $\mathbb{N}$, then every positive integer belongs to some $A_n$, so their union is $\mathbb{N}$. If their indices are bounded, they have a largest value $m$ (a nonempty bounded subset of $\mathbb{N}$ has a maximum), and the union is $A_m$. In every case the union belongs to $\mathcal{T}_1$. Hence $\mathcal{T}_1$ is a topology on $\mathbb{N}$.

**(b) The collection $\mathcal{T}_2$ is a topology.** Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_2$. For finite intersections, De Morgan's law gives

$$B_m \cap B_n = (\mathbb{N} \setminus A_m) \cap (\mathbb{N} \setminus A_n) = \mathbb{N} \setminus (A_m \cup A_n) = B_{\max\{m,n\}}.$$

Intersections involving $\varnothing$ or $\mathbb{N}$ also remain in $\mathcal{T}_2$.

Consider an arbitrary family $\{U_i\}_{i \in I} \subseteq \mathcal{T}_2$. If one member is $\mathbb{N}$, its union is $\mathbb{N}$. If every member is empty, its union is empty. Otherwise, discard the empty members and write the remaining members as $B_{n_i}$. The set of indices $\{n_i : i \in I,\ U_i \neq \varnothing\}$ is a nonempty subset of $\mathbb{N}$, so it has a least element, say $m$. Since the sets $B_n$ increase as $n$ decreases, we have

$$\bigcup_{i \in I} B_{n_i} = B_m.$$

Indeed, $B_m$ is one of the sets in the union and contains every $B_{n_i}$ because $m \leq n_i$. Thus the union belongs to $\mathcal{T}_2$. Therefore $\mathcal{T}_2$ is also a topology on $\mathbb{N}$.

**Conclusion:** Both $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

## Problem 1.7

Show that if $\mathscr{B}$ is a basis for a topology on $X$, then the topology generated by $\mathscr{B}$ equals the intersection of all topologies on $X$ that contain $\mathscr{B}$.

### Solution

*submitted by* **Sahil**

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Let $\mathcal{B}$ be a basis for a topology on $X$, and let $\mathcal{T}_{\mathcal{B}}$ denote the topology generated by $\mathcal{B}$. By the definition of the topology generated by a basis,

$$\mathcal{T}_{\mathcal{B}} = \Big\{ \bigcup_{\alpha \in A} B_\alpha : A \text{ is an index set and } B_\alpha \in \mathcal{B} \Big\}.$$

Let $\mathfrak{F}$ be the collection of all topologies on $X$ that contain $\mathcal{B}$. We prove

$$\boxed{\mathcal{T}_{\mathcal{B}} = \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.}$$

First, let $U \in \mathcal{T}_{\mathcal{B}}$. Then $U$ is a union of elements of $\mathcal{B}$. If $\mathcal{T} \in \mathfrak{F}$, then $\mathcal{B} \subseteq \mathcal{T}$. Since every topology is closed under arbitrary unions, it follows that $U \in \mathcal{T}$. This holds for every $\mathcal{T} \in \mathfrak{F}$, and hence

$$\mathcal{T}_{\mathcal{B}} \subseteq \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.$$

For the reverse inclusion, note that $\mathcal{T}_{\mathcal{B}}$ is itself a topology on $X$ containing $\mathcal{B}$. Consequently, $\mathcal{T}_{\mathcal{B}} \in \mathfrak{F}$. The intersection of all members of $\mathfrak{F}$ is therefore contained in this particular member:

$$\bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T} \subseteq \mathcal{T}_{\mathcal{B}}.$$

Combining the two inclusions proves the required equality.

**Interpretation:** The topology generated by a basis is the smallest topology on $X$ that contains every basis element.

## Problem 1.8

Let $X$ be an infinite set and $\mathcal{T}_f$ be the finite complement topology on $X$. If $U$ and $V$ are non-empty open sets in $(X, \mathcal{T}_f)$, prove that $U \cap V \neq \emptyset$.

**Solution** *submitted by* **Sahil**

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Let $X$ be infinite and let

$$\mathcal{T}_f = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}$$

be the finite-complement topology. Suppose $U, V \in \mathcal{T}_f$ are nonempty. By the definition of $\mathcal{T}_f$, both complements $X \setminus U$ and $X \setminus V$ are finite. Using De Morgan's law,

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V).$$

The right-hand side is a union of two finite sets, and is therefore finite. If $U \cap V = \varnothing$, then the left-hand side equals $X$, which would imply that $X$ is finite. This contradicts the assumption that $X$ is infinite. Hence

$$\boxed{U \cap V \neq \varnothing.}$$

## Problem 1.9

Let $X \neq \emptyset$ be a set and let $a \in X$ be a fixed point. Prove that the *particular point* collection

$$\mathcal{T}_a = \{V \subseteq X \mid V = \emptyset \text{ or } a \in V\}$$

defines a valid topology on $X$.

**Solution** *submitted by* **Sahil**

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Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a = \{V \subseteq X : V = \varnothing \text{ or } a \in V\}.$$

We check the topology axioms.

**(i) Empty set and whole space.** By definition, $\varnothing \in \mathcal{T}_a$. Also $a \in X$, so $X \in \mathcal{T}_a$.

**(ii) Arbitrary unions.** Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a$. If every $V_i$ is empty, then $\bigcup_{i \in I} V_i = \varnothing$. Otherwise, at least one member, say $V_j$, is nonempty. Since $V_j \in \mathcal{T}_a$, it contains $a$. Therefore $a \in \bigcup_{i \in I} V_i$, so the union belongs to $\mathcal{T}_a$.

**(iii) Finite intersections.** It is enough to check intersections of two members, since repeated application gives the result for every finite intersection. Let $U, V \in \mathcal{T}_a$. If either set is empty, then $U \cap V = \varnothing \in \mathcal{T}_a$. If neither is empty, both contain $a$, and hence $a \in U \cap V$. Thus $U \cap V \in \mathcal{T}_a$.

All three axioms hold. Therefore $\boxed{\mathcal{T}_a \text{ is a topology on } X.}$

## Problem 1.10

Let $X \neq \emptyset$ be a set and let $a \in X$ be a fixed point. Prove that the *excluded point* collection

$$\mathcal{T}^a := \{V \subseteq X \mid V = X \text{ or } a \notin V\}$$

defines a valid topology on $X$.

### Solution

*submitted by* **Sahil**

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Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a^e = \{V \subseteq X : V = X \text{ or } a \notin V\}.$$

Thus the open sets are $X$ itself and all subsets of $X$ that do not contain $a$. We verify the topology axioms.

**(i) Empty set and whole space.** Since $a \notin \varnothing$, we have $\varnothing \in \mathcal{T}_a^e$. Also $X \in \mathcal{T}_a^e$ by definition.

**(ii) Arbitrary unions.** Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a^e$. If some $V_j = X$, then $\bigcup_{i \in I} V_i = X$. If no member equals $X$, then every $V_i$ omits $a$. Consequently, their union also omits $a$:

$$a \notin V_i \text{ for all } i \in I \quad \Longrightarrow \quad a \notin \bigcup_{i \in I} V_i.$$

In either case the union belongs to $\mathcal{T}_a^e$.

**(iii) Finite intersections.** Again it is enough to consider two sets. Let $U, V \in \mathcal{T}_a^e$. If both equal $X$, then $U \cap V = X$. Otherwise, at least one of them, say $U$, is not $X$. By the definition of the collection, $a \notin U$. Since $U \cap V \subseteq U$, the intersection also omits $a$. Hence $U \cap V \in \mathcal{T}_a^e$.

All three topology axioms are satisfied. Therefore

$$\boxed{\mathcal{T}_a^e \text{ is a topology on } X.}$$

## Problem 1.11

Let $X$ be an infinite set and let $a \in X$ be a fixed point. Define a collection of subsets $\mathcal{T}_a \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T}_a = \mathcal{P}(X \setminus \{a\}) \cup \{V \subseteq X \mid a \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Prove that $\mathcal{T}_a$ defines a valid topology on $X$. What happens when $X$ is a finite set?

**Solution pending**

## Problem 1.12

Let $X$ be an infinite set and let $a, b \in X$ be two distinct points. Define a collection of subsets $\mathcal{T} \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T} = \mathcal{P}(X \setminus \{a, b\}) \cup \{V \subseteq X \mid a \in V \text{ or } b \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Is $\mathcal{T}$ a topology on $X$?

**Solution pending**

## Problem 1.13

Let $\mathscr{B}_1$, $\mathscr{B}_2$, and $\mathscr{B}_3$ be respectively the collections of all open balls, open rectangles, and open triangles in the plane $\mathbb{R}^2$.

(i). Prove that all three collections satisfy the basis axioms on $\mathbb{R}^2$.

(ii). Let $\mathcal{T}_i$ be the topology generated by the basis $\mathscr{B}_i$ for $i = 1, 2, 3$. Let $f\colon (\mathbb{R}^2, \mathcal{T}_i) \to (\mathbb{R}^2, \mathcal{T}_j)$ be the identity map. Prove that for any choice of $i, j \in \{1, 2, 3\}$, the map $f$ is continuous.

(iii). Conclude that $\mathcal{T}_1 = \mathcal{T}_2 = \mathcal{T}_3$.

**Solution pending**

## Problem 1.14

Along with the finite complement topology ($\mathcal{T}_f$) and the countable complement topology ($\mathcal{T}_c$), we consider the topologies generated by the following six bases:

(i) **Standard Topology ($\mathcal{T}_{\text{std}}$):** Generated by the basis of all open intervals:

$$\mathscr{B}_1 = \{(a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(ii) **$K$-Topology ($\mathcal{T}_K$):** Generated by the basis:

$$\mathscr{B}_2 = \{(a, b) \mid a < b\} \cup \{(a, b) \setminus K \mid a < b\}, \quad \text{where } K = \{1, \tfrac{1}{2}, \tfrac{1}{3}, \dots\}.$$

(iii) **Lower Limit Topology ($\mathcal{T}_\ell$):** Generated by the basis of all half-open intervals closed on the left:

$$\mathscr{B}_3 = \{[a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(iv) **Upper Limit Topology ($\mathcal{T}_u$):** Generated by the basis of all half-open intervals closed on the right:

$$\mathscr{B}_4 = \{(a, b] \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(v) **Left-Ray Topology ($\mathcal{T}_{\text{left}}$):** Generated by the basis of open left rays:

$$\mathscr{B}_5 = \{(-\infty, b) \mid b \in \mathbb{R}\}.$$

(vi) **Right-Ray Topology ($\mathcal{T}_{\text{right}}$):** Generated by the basis of open right rays:

$$\mathscr{B}_6 = \{(a, \infty) \mid a \in \mathbb{R}\}.$$

For each pair of these topologies, determine whether one contains the other (i.e., whether one is finer, coarser, or if they are completely incomparable). Summarize your results by drawing a partial order hierarchy diagram representing the containment relations among all eight topologies.

**Solution pending**

## Problem 1.15

Let $\mathcal{T}_X$ denote the collection of all topologies on a non-empty set $X$ and let $\mathcal{T}_X/\text{hom}$ be the set of homeomorphism classes of topologies on $X$. Let $S_X$ be the group of all bijective maps on $X$. Note that each bijection $f\colon X \to X$ induces a bijection $\tilde{f}\colon \mathcal{P}(X) \to \mathcal{P}(X)$ defined by $\tilde{f}(A) = \{f(a) \mid a \in A\}$ for every subset $A \in \mathcal{P}(X)$. Consequently, $S_X$ acts naturally on $\mathcal{T}_X$ via $f.\mathcal{T} = \tilde{f}(\mathcal{T})$ for all $f \in S_X$ and $\mathcal{T} \in \mathcal{T}_X$. Prove that there is a natural bijection between $\mathcal{T}_X/\text{hom}$ and the collection of orbits of the natural action of $S_X$ on $\mathcal{T}_X$.

### Solution

*submitted by* **Manas Singh**

Let $\mathcal{T}_X$ denote the collection of all topologies on $X$, and let $\mathcal{T}_X/\text{hom}$ denote the set of homeomorphism classes of topologies on $X$, where $\mathcal{T}_1 \sim \mathcal{T}_2$ if and only if $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are homeomorphic. Let $S_X$ be the group of all bijections $f\colon X \to X$. Each $f \in S_X$ induces the bijection $\tilde{f}\colon \mathcal{P}(X) \to \mathcal{P}(X)$, $\tilde{f}(A) = \{f(a) \mid a \in A\}$, and $\tilde{f}$ preserves arbitrary unions and intersections, with $\tilde{f}(\emptyset) = \emptyset$ and $\tilde{f}(X) = X$.

**The action is well-defined.** For $\mathcal{T} \in \mathcal{T}_X$, the collection $f \cdot \mathcal{T} = \tilde{f}(\mathcal{T}) = \{f(U) \mid U \in \mathcal{T}\}$ contains $\emptyset$ and $X$ and is closed under arbitrary unions and finite intersections, because $\tilde{f}$ preserves these operations. Hence $f \cdot \mathcal{T} \in \mathcal{T}_X$. Moreover $\text{id} \cdot \mathcal{T} = \mathcal{T}$ and $(g \circ f) \cdot \mathcal{T} = g \cdot (f \cdot \mathcal{T})$, so this is a group action of $S_X$ on $\mathcal{T}_X$.

**Key observation.** For $f \in S_X$ and $\mathcal{T}_1, \mathcal{T}_2 \in \mathcal{T}_X$,

$$f \cdot \mathcal{T}_1 = \mathcal{T}_2 \iff f\colon (X, \mathcal{T}_1) \to (X, \mathcal{T}_2) \text{ is a homeomorphism.}$$

Indeed, $f$ is a bijection, and $f \cdot \mathcal{T}_1 = \mathcal{T}_2$ means precisely that $U \in \mathcal{T}_1 \iff f(U) \in \mathcal{T}_2$. The forward direction ($U \in \mathcal{T}_1 \Rightarrow f(U) \in \mathcal{T}_2$) says $f^{-1}$ is continuous, and the backward direction says $f$ is continuous. So $f$ is a homeomorphism.

Conversely, every homeomorphism $f\colon (X, \mathcal{T}_1) \to (X, \mathcal{T}_2)$ is in particular a bijection $X \to X$, i.e. an element of $S_X$, and it satisfies $f \cdot \mathcal{T}_1 = \mathcal{T}_2$.

**The bijection.** Define $\Phi\colon \mathcal{T}_X/S_X \to \mathcal{T}_X/\text{hom}$ by

$$\Phi(\text{Orb}(\mathcal{T})) = [\mathcal{T}]_{\text{hom}}.$$

- **Well-defined and injective:** By the key observation,

$$\begin{aligned} \text{Orb}(\mathcal{T}_1) = \text{Orb}(\mathcal{T}_2) &\iff \exists f \in S_X :\ f \cdot \mathcal{T}_1 = \mathcal{T}_2 \\ &\iff (X, \mathcal{T}_1) \cong (X, \mathcal{T}_2) \iff [\mathcal{T}_1]_{\text{hom}} = [\mathcal{T}_2]_{\text{hom}}. \end{aligned}$$

  The implication "$\Rightarrow$" shows that $\Phi$ does not depend on the choice of representative of the orbit (well-defined), and "$\Leftarrow$" shows that $\Phi$ is injective.

- **Surjective:** Given any class $[\mathcal{T}]_{\text{hom}} \in \mathcal{T}_X/\text{hom}$, we have $\Phi(\text{Orb}(\mathcal{T})) = [\mathcal{T}]_{\text{hom}}$.

Therefore $\Phi$ is a natural bijection (it requires no choices) between the orbits of the action of $S_X$ on $\mathcal{T}_X$ and $\mathcal{T}_X/\text{hom}$.

## Problem 1.16

List down all possible (there are 29) topologies on a three-point set $X = \{a, b, c\}$. Compute the number of orbits under the natural action of $S_3$ on the set $\mathcal{T}_X$ of all topologies on $X$.

### Solution

*submitted by* **Manas Singh**

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Let $X = \{a, b, c\}$. The group $S_3$ acts on $\mathcal{T}_X$ by $f \cdot \mathcal{T} = \{f(U) \mid U \in \mathcal{T}\}$. Every topology contains $\emptyset$ and $X$, so we list all 29 topologies explicitly, grouped by orbit. Within each orbit, the topologies are obtained from one another by permuting $a, b, c$.

1. **Indiscrete topology** (orbit size 1):
   - $\{\emptyset, X\}$
2. **One non-trivial open set, a singleton** (orbit size 3):
   - $\{\emptyset, \{a\}, X\}$
   - $\{\emptyset, \{b\}, X\}$
   - $\{\emptyset, \{c\}, X\}$
3. **One non-trivial open set, a two-element set** (orbit size 3):
   - $\{\emptyset, \{a, b\}, X\}$
   - $\{\emptyset, \{b, c\}, X\}$
   - $\{\emptyset, \{a, c\}, X\}$
4. **Partition topology** (orbit size 3):
   - $\{\emptyset, \{a\}, \{b, c\}, X\}$
   - $\{\emptyset, \{b\}, \{a, c\}, X\}$
   - $\{\emptyset, \{c\}, \{a, b\}, X\}$
5. **Nested chain** $\{x\} \subset \{x, y\} \subset X$ (orbit size 6, one for each ordered pair $(x, y)$ of distinct points):
   - $\{\emptyset, \{a\}, \{a, b\}, X\}$
   - $\{\emptyset, \{a\}, \{a, c\}, X\}$
   - $\{\emptyset, \{b\}, \{a, b\}, X\}$
   - $\{\emptyset, \{b\}, \{b, c\}, X\}$
   - $\{\emptyset, \{c\}, \{a, c\}, X\}$
   - $\{\emptyset, \{c\}, \{b, c\}, X\}$
6. **Two disjoint singletons and their union** (orbit size 3, determined by the excluded point):
   - $\{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$
   - $\{\emptyset, \{b\}, \{c\}, \{b, c\}, X\}$

- $\{\emptyset, \{a\}, \{c\}, \{a, c\}, X\}$

7. **One singleton with both two-element sets containing it** (orbit size 3, determined by the singleton):
   - $\{\emptyset, \{a\}, \{a, b\}, \{a, c\}, X\}$
   - $\{\emptyset, \{b\}, \{a, b\}, \{b, c\}, X\}$
   - $\{\emptyset, \{c\}, \{a, c\}, \{b, c\}, X\}$
8. **Two singletons $\{x\}, \{y\}$ with $\{x, y\}$ and $\{x, z\}$** (orbit size 6, one for each ordered pair $(x, y)$ of distinct points, where $z$ is the remaining point):
   - $\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{a, c\}, X\}$
   - $\{\emptyset, \{b\}, \{a\}, \{a, b\}, \{b, c\}, X\}$
   - $\{\emptyset, \{a\}, \{c\}, \{a, c\}, \{a, b\}, X\}$
   - $\{\emptyset, \{c\}, \{a\}, \{a, c\}, \{b, c\}, X\}$
   - $\{\emptyset, \{b\}, \{c\}, \{b, c\}, \{a, b\}, X\}$
   - $\{\emptyset, \{c\}, \{b\}, \{b, c\}, \{a, c\}, X\}$
9. **Discrete topology** (orbit size 1):
   - $\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, X\}$

Each listed collection contains $\emptyset$ and $X$ and is closed under unions and intersections, so each is a topology. Types 1–9 are pairwise non-homeomorphic: they differ in the number of open sets $(2, 3, 3, 4, 4, 5, 5, 6, 8)$ or, when these agree, in their structure (types 2 and 3 differ in the size of the non-trivial open set; types 4 and 5 differ because the open sets of type 4 are disjoint while those of type 5 are nested; types 6 and 7 differ because type 6 has two singleton open sets and type 7 has only one). Within each type, permuting the points moves any topology to any other, so each type is exactly one orbit.

Summing the orbit sizes gives

$$1 + 3 + 3 + 3 + 6 + 3 + 3 + 6 + 1 = 29,$$

which accounts for all 29 topologies. Hence the number of orbits of $S_3$ on $\mathcal{T}_X$, i.e. the number of homeomorphism classes of topologies on a three-point set, is exactly **9**.

## Problem 1.17

Let $X$ be a set, and let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two subbases on $X$ generating topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively.

(i). If $\mathcal{S}_1 \subseteq \mathcal{S}_2$, prove that $\mathcal{T}_1 \subseteq \mathcal{T}_2$.

(ii). Give an explicit example where $\mathcal{T}_1 = \mathcal{T}_2$ even though $\mathcal{S}_1 \neq \mathcal{S}_2$.

### Solution

*submitted by* **Manas Singh**

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Recall that the basis generated by a subbasis $\mathcal{S}$ is $\mathscr{B}_{\mathcal{S}} = \{U_1 \cap \cdots \cap U_n \mid n \in \mathbb{N},\ U_i \in \mathcal{S}\}$, and the topology generated by $\mathcal{S}$ consists of all unions of elements of $\mathscr{B}_{\mathcal{S}}$.

**(i)** Suppose $\mathcal{S}_1 \subseteq \mathcal{S}_2$. Let $W \in \mathcal{T}_1$. Then $W = \bigcup_{\alpha \in I} B_\alpha$, where each $B_\alpha$ is a finite intersection of elements of $\mathcal{S}_1$. Since $\mathcal{S}_1 \subseteq \mathcal{S}_2$, each set in that finite intersection belongs to $\mathcal{S}_2$, so $B_\alpha \in \mathscr{B}_{\mathcal{S}_2}$ for all $\alpha \in I$. Thus $W$ is a union of elements of $\mathscr{B}_{\mathcal{S}_2}$, i.e. $W \in \mathcal{T}_2$. Therefore $\mathcal{T}_1 \subseteq \mathcal{T}_2$.

**(ii)** We give an explicit example where $\mathcal{T}_1 = \mathcal{T}_2$ even though $\mathcal{S}_1 \neq \mathcal{S}_2$.

Let $X = \{1, 2, 3\}$ and define

$$\mathcal{S}_1 = \{\{1\}, \{2\}, X\}, \qquad \mathcal{S}_2 = \{\{1\}, \{2\}, \{1, 2\}, X\}.$$

Both cover $X$ (they contain $X$), so both are subbases. Clearly $\mathcal{S}_1 \neq \mathcal{S}_2$, since $\{1, 2\} \in \mathcal{S}_2 \setminus \mathcal{S}_1$.

- The finite intersections of elements of $\mathcal{S}_1$ are $\{1\}$, $\{2\}$, $X$ and $\{1\} \cap \{2\} = \emptyset$, so $\mathscr{B}_1 = \{\emptyset, \{1\}, \{2\}, X\}$. Taking all unions (note $\{1\} \cup \{2\} = \{1, 2\}$),

$$\mathcal{T}_1 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}.$$

- The finite intersections of elements of $\mathcal{S}_2$ are $\{1\}, \{2\}, \{1, 2\}, X$ and $\emptyset$ (for instance $\{1\} \cap \{2\}$), and no new sets arise from intersecting these (e.g. $\{1, 2\} \cap \{1\} = \{1\}$). So $\mathscr{B}_2 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$, and since this family is already closed under unions,

$$\mathcal{T}_2 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}.$$

Thus $\mathcal{T}_1 = \mathcal{T}_2$ although $\mathcal{S}_1 \neq \mathcal{S}_2$.

# Problem Set 2

## Problem 2.1

Whether each of the following subsets of $\mathbb{R}^2$ can be written as a Cartesian product $A \times B$, where $A, B \subset \mathbb{R}$?

(i) $\{(x, y) \mid x \text{ is an integer}\}$.

(ii) $\{(x, y) \mid 0 \leq x < 1\}$.

(iii) $\{(x, y) \mid x < y\}$.

(iv) $\{(x, y) \mid x \text{ is an integer, and } y \text{ is not a rational number}\}$.

(v) $\{(x, y) \mid x^2 + y^2 \leq 2026\}$.

**Solution** *submitted by* **Harisankar B**

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We use the following observation repeatedly: if $S = A \times B$ and $(a, b), (a', b') \in S$, then $a, a' \in A$ and $b, b' \in B$, so also $(a, b'), (a', b) \in S$.

(i) **Yes.** $\{(x, y) \mid x \in \mathbb{Z}\} = \mathbb{Z} \times \mathbb{R}$.

(ii) **Yes.** $\{(x, y) \mid 0 \leq x < 1\} = [0, 1) \times \mathbb{R}$.

(iii) **No.** Let $S = \{(x, y) \mid x < y\}$ and suppose $S = A \times B$. Since $(0, 1) \in S$ we get $1 \in B$, and since $(1, 2) \in S$ we get $1 \in A$. Hence $(1, 1) \in A \times B = S$, i.e. $1 < 1$, a contradiction.

(iv) **Yes.** $\{(x, y) \mid x \in \mathbb{Z},\ y \notin \mathbb{Q}\} = \mathbb{Z} \times (\mathbb{R} \setminus \mathbb{Q})$.

(v) **No.** Let $S = \{(x, y) \mid x^2 + y^2 \leq 2026\}$, put $r = \sqrt{2026}$, and suppose $S = A \times B$. Since $(r, 0) \in S$ we get $r \in A$, and since $(0, r) \in S$ we get $r \in B$. Hence $(r, r) \in S$, i.e. $2r^2 = 4052 \leq 2026$, a contradiction.

## Problem 2.2

Let $f \colon A \to B$ be a function, with $A_0 \subseteq A$ and $B_0 \subseteq B$.

(i) Show that $A_0 \subseteq f^{-1}(f(A_0))$, and that equality holds for every subset $A_0 \subseteq A$ if and only if $f$ is injective.

(ii) Show that $f(f^{-1}(B_0)) \subseteq B_0$, and that equality holds for every subset $B_0 \subseteq B$ if and only if $f$ is surjective.

**Solution** *submitted by* **Harisankar B**

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**(i)** Let $a \in A_0$. Then $f(a) \in f(A_0)$, so $a \in f^{-1}(f(A_0))$. Hence $A_0 \subseteq f^{-1}(f(A_0))$.

Suppose $f$ is injective and let $x \in f^{-1}(f(A_0))$. Then $f(x) \in f(A_0)$, so $f(x) = f(a)$ for some $a \in A_0$. By injectivity $x = a \in A_0$. Thus $f^{-1}(f(A_0)) \subseteq A_0$, and equality holds for every $A_0 \subseteq A$.

Conversely, suppose equality holds for every $A_0 \subseteq A$, and let $f(x) = f(y)$. Taking $A_0 = \{x\}$, we have $f(y) = f(x) \in f(\{x\})$, so $y \in f^{-1}(f(\{x\})) = \{x\}$, i.e. $y = x$. Hence $f$ is injective.

**(ii)** Let $y \in f(f^{-1}(B_0))$. Then $y = f(x)$ for some $x \in f^{-1}(B_0)$, i.e. $f(x) \in B_0$. Hence $y \in B_0$, and $f(f^{-1}(B_0)) \subseteq B_0$.

Suppose $f$ is surjective and let $y \in B_0$. Choose $x \in A$ with $f(x) = y$. Then $f(x) \in B_0$, so $x \in f^{-1}(B_0)$ and $y = f(x) \in f(f^{-1}(B_0))$. Thus $B_0 \subseteq f(f^{-1}(B_0))$, and equality holds for every $B_0 \subseteq B$.

Conversely, suppose equality holds for every $B_0 \subseteq B$. Taking $B_0 = B$ and using $f^{-1}(B) = A$, we get $f(A) = f(f^{-1}(B)) = B$. Hence $f$ is surjective.

## Problem 2.3

Let $f\colon A \to B$ be a function, and $B_0, B_1 \subseteq B$. Show that preimages preserve inclusions, unions, intersections, and set differences by proving the following:

(i) If $B_0 \subseteq B_1$, then $f^{-1}(B_0) \subseteq f^{-1}(B_1)$.

(ii) $f^{-1}(B_0 \cup B_1) = f^{-1}(B_0) \cup f^{-1}(B_1)$.

(iii) $f^{-1}(B_0 \cap B_1) = f^{-1}(B_0) \cap f^{-1}(B_1)$.

(iv) $f^{-1}(B_0 \setminus B_1) = f^{-1}(B_0) \setminus f^{-1}(B_1)$.

### Solution

*submitted by* **Harisankar B**

Throughout, recall that $x \in f^{-1}(C)$ if and only if $f(x) \in C$.

**(i)** Let $B_0 \subseteq B_1$ and $x \in f^{-1}(B_0)$. Then $f(x) \in B_0 \subseteq B_1$, so $x \in f^{-1}(B_1)$. Hence $f^{-1}(B_0) \subseteq f^{-1}(B_1)$.

**(ii)** For $x \in A$,

$$\begin{aligned} x \in f^{-1}(B_0 \cup B_1) &\iff f(x) \in B_0 \text{ or } f(x) \in B_1 \\ &\iff x \in f^{-1}(B_0) \text{ or } x \in f^{-1}(B_1) \iff x \in f^{-1}(B_0) \cup f^{-1}(B_1). \end{aligned}$$

**(iii)** For $x \in A$,

$$\begin{aligned} x \in f^{-1}(B_0 \cap B_1) &\iff f(x) \in B_0 \text{ and } f(x) \in B_1 \\ &\iff x \in f^{-1}(B_0) \text{ and } x \in f^{-1}(B_1) \iff x \in f^{-1}(B_0) \cap f^{-1}(B_1). \end{aligned}$$

**(iv)** For $x \in A$,

$$\begin{aligned} x \in f^{-1}(B_0 \setminus B_1) &\iff f(x) \in B_0 \text{ and } f(x) \notin B_1 \\ &\iff x \in f^{-1}(B_0) \text{ and } x \notin f^{-1}(B_1) \iff x \in f^{-1}(B_0) \setminus f^{-1}(B_1). \end{aligned}$$

In each of (ii)–(iv) the two sets have the same elements, so they are equal.

## Problem 2.4

Let $f\colon A \to B$ be a function, and let $A_0, A_1 \subseteq A$.

(i) Show that images preserve inclusions and unions:

(a) If $A_0 \subseteq A_1$, then $f(A_0) \subseteq f(A_1)$.

(b) $f(A_0 \cup A_1) = f(A_0) \cup f(A_1)$.

(ii) Show that images satisfy only one-way inclusions for intersections and set differences in general:

(a) $f(A_0 \cap A_1) \subseteq f(A_0) \cap f(A_1)$.

(b) $f(A_0) \setminus f(A_1) \subseteq f(A_0 \setminus A_1)$.

(iii) Prove that equality holds in both (ii)(a) and (ii)(b) for all subsets $A_0, A_1 \subseteq A$ if and only if $f$ is injective.

### Solution

*submitted by* **Harisankar B**

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**(i)(a)** Let $A_0 \subseteq A_1$ and $y \in f(A_0)$. Then $y = f(a)$ with $a \in A_0 \subseteq A_1$, so $y \in f(A_1)$.

**(i)(b)** We have

$$\begin{aligned} y \in f(A_0 \cup A_1) &\iff y = f(a) \text{ for some } a \in A_0 \text{ or some } a \in A_1 \\ &\iff y \in f(A_0) \text{ or } y \in f(A_1) \iff y \in f(A_0) \cup f(A_1). \end{aligned}$$

**(ii)(a)** Since $A_0 \cap A_1 \subseteq A_0$ and $A_0 \cap A_1 \subseteq A_1$, part (i)(a) gives $f(A_0 \cap A_1) \subseteq f(A_0)$ and $f(A_0 \cap A_1) \subseteq f(A_1)$, hence $f(A_0 \cap A_1) \subseteq f(A_0) \cap f(A_1)$.

**(ii)(b)** Let $y \in f(A_0) \setminus f(A_1)$. Then $y = f(a)$ for some $a \in A_0$. If $a \in A_1$ we would have $y \in f(A_1)$, which is false; so $a \in A_0 \setminus A_1$ and $y \in f(A_0 \setminus A_1)$.

*The reverse inclusions fail in general.* Let $f\colon \{1,2\} \to \{0\}$ be the constant map, $A_0 = \{1\}$, $A_1 = \{2\}$. Then $f(A_0 \cap A_1) = f(\emptyset) = \emptyset$, while $f(A_0) \cap f(A_1) = \{0\}$; and $f(A_0) \setminus f(A_1) = \emptyset$, while $f(A_0 \setminus A_1) = f(\{1\}) = \{0\}$.

**(iii)** Suppose $f$ is injective.

- Let $y \in f(A_0) \cap f(A_1)$, say $y = f(a_0) = f(a_1)$ with $a_0 \in A_0$, $a_1 \in A_1$. Injectivity gives $a_0 = a_1 \in A_0 \cap A_1$, so $y \in f(A_0 \cap A_1)$. With (ii)(a), $f(A_0 \cap A_1) = f(A_0) \cap f(A_1)$.
- Let $y \in f(A_0 \setminus A_1)$, say $y = f(a)$ with $a \in A_0$, $a \notin A_1$. If $y \in f(A_1)$, then $y = f(a')$ for some $a' \in A_1$, and injectivity gives $a = a' \in A_1$, a contradiction. So $y \in f(A_0) \setminus f(A_1)$. With (ii)(b), $f(A_0) \setminus f(A_1) = f(A_0 \setminus A_1)$.

Conversely, suppose equality holds in (ii)(a) and (ii)(b) for all $A_0, A_1 \subseteq A$, and suppose $f(x) = f(x')$ with $x \neq x'$. Take $A_0 = \{x\}$, $A_1 = \{x'\}$. Then $f(A_0 \cap A_1) = f(\emptyset) = \emptyset$, but $f(A_0) \cap f(A_1) = \{f(x)\} \neq \emptyset$, contradicting equality in (ii)(a). Hence $f$ is injective.

## Problem 2.5

Let $\sim$ be a relation defined on the Euclidean plane $\mathbb{R}^2$ as follows: two points are related if and only if their distances from the origin are equal.

(i) Show that $\sim$ is an equivalence relation (i.e., reflexivity, symmetry, and transitivity).

(ii) Describe the equivalence class $[P]$ for any point $P \in \mathbb{R}^2$.

(iii) What is the geometric interpretation of the partition of $\mathbb{R}^2$ induced by this relation?

### Solution

*submitted by* **Harisankar B**

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For $P = (x, y) \in \mathbb{R}^2$ write $\|P\| = \sqrt{x^2 + y^2}$ for its distance from the origin $O$. Thus $P \sim Q$ if and only if $\|P\| = \|Q\|$.

**(i)** Since $\sim$ is defined by equality of the real numbers $\|\cdot\|$, it inherits the properties of equality:

- *Reflexive:* $\|P\| = \|P\|$, so $P \sim P$.
- *Symmetric:* if $\|P\| = \|Q\|$ then $\|Q\| = \|P\|$, so $P \sim Q \Rightarrow Q \sim P$.
- *Transitive:* if $\|P\| = \|Q\|$ and $\|Q\| = \|R\|$ then $\|P\| = \|R\|$, so $P \sim Q$, $Q \sim R \Rightarrow P \sim R$.

Hence $\sim$ is an equivalence relation.

**(ii)** Let $r = \|P\| \geq 0$. Then

$$[P] = \{Q \in \mathbb{R}^2 \mid \|Q\| = r\} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = r^2\}.$$

If $P \neq O$ (so $r > 0$), $[P]$ is the circle of radius $r$ centred at the origin passing through $P$. If $P = O$, then $[P] = \{O\}$, a single point (a "circle of radius 0").

**(iii)** The equivalence classes partition $\mathbb{R}^2$ into the origin together with the family of concentric circles $\{x^2 + y^2 = r^2\}$, $r > 0$, centred at the origin. Every point of the plane lies on exactly one such circle (the one of radius equal to its distance from $O$), and distinct radii give disjoint circles. The map $[P] \mapsto \|P\|$ is a bijection from the set of equivalence classes $\mathbb{R}^2/\sim$ onto $[0, \infty)$, so the classes are labelled exactly by the possible radii.

## Problem 2.6

Let $X$ be a non-empty set. Suppose that $S$ is the collection of the following two types of open rays: $(a, \infty) = \{x \in X \mid a < x\}$, and $(-\infty, b) = \{x \in X \mid x < b\}$. Prove that $S$ is a subbasis, and it generates the order topology on $X$.

### Solution

*submitted by* **Kanishk**

Let $S$ be the collection of sets of the form

$$(a, \infty) = \{x \in X : a < x\} \qquad \text{and} \qquad (-\infty, b) = \{x \in X : x < b\}.$$

For two rays of the same type, we have

$$(a, \infty) \cap (c, \infty) = (\max\{a, c\}, \infty) \qquad \text{and} \qquad (-\infty, b) \cap (-\infty, d) = (-\infty, \min\{b, d\}).$$

Also, for two rays of opposite types,

$$(a, \infty) \cap (-\infty, b) = (a, b).$$

Thus, finite intersections of elements of $S$ give open intervals and open rays, which form a basis for the order topology. Hence, $S$ is a subbasis and generates the order topology on $X$.

## Problem 2.7

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of non-empty countable sets.

(1) Prove that the finite Cartesian product $\prod_{i=1}^{k} A_i = A_1 \times A_2 \times \cdots \times A_k$ is countable for any $k \in \mathbb{N}$.

(2) Determine whether the infinite Cartesian product $\prod_{n=1}^{\infty} A_n$ is necessarily countable. Justify your answer.

### Solution

*submitted by* **Kanishk**

**(1)** We prove this by induction on $k$.

For $k = 1$, the result is clear since $A_1$ is countable.

Suppose that $A_1 \times A_2 \times \cdots \times A_k$ is countable. Since $A_{k+1}$ is also countable and the product of two countable sets is countable, we get that

$$(A_1 \times \cdots \times A_k) \times A_{k+1}$$

is countable. Therefore, by induction, $\prod_{i=1}^{k} A_i$ is countable for every $k \in \mathbb{N}$.

**(2)** No, the infinite Cartesian product is not necessarily countable. Consider

$$A_n = \{0, 1\}, \qquad n \in \mathbb{N}.$$

Suppose that $\prod_{n=1}^{\infty} A_n$ is countable. Then we can list all its elements as

$$x_1, x_2, x_3, \ldots, \qquad \text{where} \qquad x_i = (x_{i1}, x_{i2}, \ldots).$$

Define a new sequence $y = (y_1, y_2, \ldots)$ by

$$y_n = 1 - x_{nn}.$$

Clearly, $y$ belongs to the infinite product, since every entry is either 0 or 1. But $y$ is different from every $x_n$, since their $n$-th entries are different.

This is a contradiction. Hence, $\prod_{n=1}^{\infty} A_n$ is not necessarily countable.

## Problem 2.8

Let $X$ be a linearly ordered set. Prove that a singleton set $\{x\}$ is open in the order topology on $X$ if and only if $x$ has an immediate predecessor and an immediate successor, or if $x$ is the minimum/maximum and has an immediate successor/predecessor respectively.

### Solution

*submitted by* **Kanishk**

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Suppose first that $\{x\}$ is open in the order topology. Then there exists a basic open set $B$ such that

$$x \in B \subseteq \{x\}.$$

Hence, $B = \{x\}$.

If $x$ is neither the minimum nor the maximum element, then $B$ must be an open interval $(a, b)$, where $a < x < b$. Since $(a, b) = \{x\}$, there is no element between $a$ and $x$, or between $x$ and $b$. Thus, $a$ is the immediate predecessor and $b$ is the immediate successor of $x$.

If $x$ is the minimum element, then

$$(-\infty, b) = \{x\},$$

so $b$ is the immediate successor of $x$. Similarly, if $x$ is the maximum element, then

$$(a, \infty) = \{x\},$$

so $a$ is the immediate predecessor of $x$.

Conversely, suppose that $x$ has an immediate predecessor $a$ and an immediate successor $b$. Then

$$(a, b) = \{x\}.$$

Since $(a, b)$ is open, $\{x\}$ is open.

If $x$ is the minimum element and has an immediate successor $b$, then

$$(-\infty, b) = \{x\},$$

which is open. Similarly, if $x$ is the maximum element and has an immediate predecessor $a$, then

$$(a, \infty) = \{x\},$$

which is open.

Therefore, the given conditions are necessary and sufficient for $\{x\}$ to be open.

## Problem 2.9

Prove that if $V$ is closed in $Y$ and $Y$ is closed in $X$, then $V$ is closed in $X$.

**Solution** *submitted by* **Karmukilan**

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*Conventions.* A subspace $Y \subseteq X$ carries the topology $\{Y \cap U : U \text{ open in } X\}$. A *neighborhood* of $x$ is an open set containing $x$. The closure $\overline{A}$ is the intersection of all closed sets containing $A$; it is the smallest closed set containing $A$.

*Proof.* Since $V$ is closed in the subspace $Y$, the set $Y \setminus V$ is open in $Y$. By definition of the subspace topology, there is an open set $U \subseteq X$ with

$$Y \setminus V = Y \cap U.$$

Taking complements inside $Y$ gives

$$V = Y \setminus (Y \setminus V) = Y \setminus (Y \cap U) = Y \cap (X \setminus U).$$

Here $X \setminus U$ is closed in $X$ because $U$ is open in $X$, and $Y$ is closed in $X$ by hypothesis. Hence $V$ is an intersection of two closed subsets of $X$, so $V$ is closed in $X$. □

## Problem 2.10

Let $Y$ be a subspace of $X$ and let $A$ be a subset of $Y$. Prove that the topology $A$ inherits as a subspace of $Y$ is identical to the topology it inherits directly as a subspace of $X$.

**Solution** *submitted by* **Karmukilan**

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Let $\tau_1$ be the topology that $A$ inherits as a subspace of $Y$, and $\tau_2$ the topology that $A$ inherits as a subspace of $X$. We show $\tau_1 = \tau_2$.

*Proof.* Explicitly,

$$\tau_1 = \{A \cap W : W \text{ open in } Y\}, \qquad \tau_2 = \{A \cap U : U \text{ open in } X\}.$$

$\tau_1 \subseteq \tau_2$**:** Let $A \cap W \in \tau_1$ with $W$ open in $Y$. Then $W = Y \cap U$ for some $U$ open in $X$. Since $A \subseteq Y$,

$$A \cap W = A \cap (Y \cap U) = (A \cap Y) \cap U = A \cap U \in \tau_2.$$

$\tau_2 \subseteq \tau_1$**:** Let $A \cap U \in \tau_2$ with $U$ open in $X$. Then $W := Y \cap U$ is open in $Y$, and since $A \subseteq Y$,

$$A \cap U = (A \cap Y) \cap U = A \cap (Y \cap U) = A \cap W \in \tau_1.$$

Therefore $\tau_1 = \tau_2$. □

## Problem 2.11

Let $A$ be a subset of a topological space $X$ and $x \in X$ be an element. Then prove that the following statements are equivalent:

(i). The point $x \in \overline{A}$.

(ii). Every neighborhood of $x$ intersects $A$.

(iii). Every basis element containing $x$ intersects $A$.

### Solution

*submitted by* **Karmukilan**

Let $\mathscr{B}$ be a basis for the topology of $X$.

*Proof.* **(i) $\Rightarrow$ (ii).** We prove the contrapositive. Suppose some neighborhood $U$ of $x$ satisfies $U \cap A = \emptyset$. Then $A \subseteq X \setminus U$, and $X \setminus U$ is closed since $U$ is open. As $\overline{A}$ is the smallest closed set containing $A$, we get $\overline{A} \subseteq X \setminus U$. Since $x \in U$, $x \notin X \setminus U$, hence $x \notin \overline{A}$.

**(ii) $\Rightarrow$ (iii).** Every basis element $B \in \mathscr{B}$ is open, so a basis element containing $x$ is a neighborhood of $x$. By (ii) it intersects $A$.

**(iii) $\Rightarrow$ (i).** Again by contrapositive. Suppose $x \notin \overline{A}$. Then $U := X \setminus \overline{A}$ is open (since $\overline{A}$ is closed) and contains $x$. Because $\mathscr{B}$ is a basis, there is $B \in \mathscr{B}$ with $x \in B \subseteq U$. Then

$$B \cap A \subseteq U \cap A \subseteq (X \setminus \overline{A}) \cap A = \emptyset,$$

since $A \subseteq \overline{A}$. So $B$ is a basis element containing $x$ that does not intersect $A$, and (iii) fails.

Thus (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i), and the three statements are equivalent. □

## Problem 2.12

Prove that $\overline{A} = A \cup A'$. Deduce that $A$ is closed in $X$ if and only if $A' \subseteq A$.

### Solution

*submitted by* **Srijit Bhadra**

*Solution:* Let $x \in \bar{A}$. Now,

$$\begin{aligned} x \in \bar{A} &\iff \text{every neighbourhood of } x \text{ intersects } A \\ &\iff x \in A \text{ or } x \in A' \\ &\iff x \in A \cup A'. \end{aligned}$$

Since $x$ was chosen arbitrarily, we can say $\bar{A} = A \cup A'$.

For the second part, assume $A$ is closed. Then $\bar{A} = A \implies A \cup A' = A \implies A' \subseteq A$. On the other hand, if we assume $A' \subseteq A$, then we get $\bar{A} = A \cup A' = A$. Thus, in this case, we can conclude that $A$ is closed. As both directions of the double implication hold, we have $A$ is closed in $X$ if and only if $A' \subseteq A$. □

## Problem 2.13

Prove that $\operatorname{Int} A \cap \operatorname{Bd} A = \emptyset$, and $\overline{A} = \operatorname{Int} A \cup \operatorname{Bd} A$.

### Solution

*submitted by* **Srijit Bhadra**

*Solution:* We know, $\operatorname{Int}(A) \cap \operatorname{Bd}(A) = \operatorname{Int}(A) \cap \bar{A} \cap \overline{X \setminus A}$. For the sake of contradiction, assume there is a point $x \in \operatorname{Int}(A) \cap \operatorname{Bd}(A) \implies x \in \operatorname{Int}(A) \cap \bar{A} \cap \overline{X \setminus A}$. As $x \in \overline{X \setminus A}$, every neighbourhood of $x$ must intersect $X \setminus A$. Thus, $\operatorname{Int}(A) \cap \overline{X \setminus A} \neq \emptyset$. But this is impossible as $\operatorname{Int}(A) \subseteq A$ ($\Rightarrow\Leftarrow$). Thus, $\operatorname{Int}(A) \cap \operatorname{Bd}(A) = \emptyset$.

For the second part, let $x \in \bar{A}$. Again, this happens if and only if every neighborhood of $x$ intersects $A$. Now, either there exists a neighborhood $U$ of $x$ such that $x \in U \subseteq A$, or there is no such neighborhood. In the first case, $x \in U \subseteq A \implies x \in \operatorname{Int}(A)$ since $\operatorname{Int}(A)$ is the largest open set contained in $A$. In the second case, every neighborhood $V$ of $x$ intersects $X \setminus A \implies x \in \overline{X \setminus A} \implies x \in \operatorname{Bd}(A)$. Combining the two cases, $x \in \operatorname{Int}(A) \cup \operatorname{Bd}(A)$. Since $x \in \bar{A}$ was chosen arbitrarily, $\bar{A} \subseteq \operatorname{Int}(A) \cup \operatorname{Bd}(A)$.

For the other way inclusion, notice that $\operatorname{Int}(A) \subseteq A \subseteq \bar{A}$. Also, $\operatorname{Bd}(A) = \bar{A} \cap \overline{X \setminus A} \subseteq \bar{A}$. Therefore, $\operatorname{Int}(A) \cup \operatorname{Bd}(A) \subseteq \bar{A}$. Hence, we conclude that $\bar{A} = \operatorname{Int}(A) \cup \operatorname{Bd}(A)$. □

## Problem 2.14

Prove that every order topology is Hausdorff.

### Solution

*submitted by* **Srijit Bhadra**

*Solution:* Let $X \neq \emptyset$ and $\mathscr{T}$ be the order topology on $X$. Let $x_1, x_2 \in X$ such that $x_1 \neq x_2$. Consider the basic open set $(x_1, x_2) = \{x \in X : x_1 < x < x_2\} \in \mathscr{T}$. Now, there can be two cases:

**Case 1.** $(x_1, x_2) = \emptyset$. Observe that

$$(-\infty, x_2) = \bigcup_{a \in X,\ a < x_2} (a, x_2) \in \mathscr{T}.$$

Similarly,

$$(x_1, \infty) = \bigcup_{b \in X,\ b > x_1} (x_1, b) \in \mathscr{T}.$$

Also, $x_1 \in (-\infty, x_2)$ and $x_2 \in (x_1, \infty)$. Crucially, $(-\infty, x_2) \cap (x_1, \infty) = \emptyset$.

**Case 2.** $(x_1, x_2) \neq \emptyset$. Take any point $c \in (x_1, x_2)$. As we saw earlier, $(-\infty, c)$ and $(c, \infty)$ are open in $X$. Note that $x_1 \in (-\infty, c)$, $x_2 \in (c, \infty)$, and $(-\infty, c) \cap (c, \infty) = \emptyset$.

Thus, in both cases, we found two disjoint open sets in $X$ with one containing $x_1$ and the other containing $x_2$. As $x_1, x_2 \in X$ were chosen arbitrarily, we can say that $(X, \mathscr{T})$ is Hausdorff. Since $(X, \mathscr{T})$ was also chosen arbitrarily, we can say that every order topology is Hausdorff. □

## Problem 2.15

Let $X$ be a $T_1$ space. Then prove that for each pair $x_1, x_2$ of distinct points, there exist open sets $U_1$ and $U_2$ containing $x_1$ and $x_2$ respectively such that $x_1 \notin U_2$ and $x_2 \notin U_1$.

### Solution

*submitted by* **Srijit Bhadra**

*Solution:* Let $x_1, x_2$ be a pair of distinct points in $X$. $X$ being a $T_1$ space, singletons $\{x_1\}$ and $\{x_2\}$ are closed in $X$. Therefore, $X \setminus \{x_1\}$ and $X \setminus \{x_2\}$ are open in $X$. Now, $x_2 \in X \setminus \{x_1\}$ (because $x_1 \neq x_2$) but $x_2 \notin X \setminus \{x_2\}$. Similarly, $x_1 \in X \setminus \{x_2\}$ (because $x_1 \neq x_2$) but $x_1 \notin X \setminus \{x_1\}$. As $x_1, x_2 \in X$ were chosen arbitrarily, we can say that for each pair $x_1, x_2$ of distinct points, there exist open sets $U_1$ and $U_2$ containing $x_1$ and $x_2$, respectively, such that $x_1 \notin U_2$ and $x_2 \notin U_1$. $\square$

# Problem Set 3

## Problem 3.1

Give an example of a $T_1$ space where a sequence converges to more than one point. Conclude that a $T_1$ space is not necessarily a Hausdorff space.

**Solution** *submitted by* **Arunodaya**

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Consider the set $X = \mathbb{R}$ equipped with the cofinite topology

$$\tau = \{\emptyset\} \cup \{U \subseteq \mathbb{R} : \mathbb{R} \setminus U \text{ is finite}\},$$

that is, the open sets are the empty set or subsets of $\mathbb{R}$ whose complements are finite.

Now, we know that a topological space is $T_1$ if every singleton set is closed. Let $x \in \mathbb{R}$. Then $\mathbb{R} \setminus \{x\}$ is open because its complement $\{x\}$ is finite. Therefore, $\{x\}$ is closed. Since $x$ is arbitrary, every singleton subset of $\mathbb{R}$ is closed. Hence $(\mathbb{R}, \tau)$ is a $T_1$ space.

Consider the sequence

$$x_n = n, \quad n \in \mathbb{N}.$$

$x_n \to x$ if for every open neighbourhood $U$ of $x$, $\exists\, N \in \mathbb{N}$ such that $n \geq N \Rightarrow x_n \in U$.

Let $x \in \mathbb{R}$ be arbitrary. Let $U$ be any open neighbourhood of $x$. By the definition of the cofinite topology, $\mathbb{R} \setminus U$ is finite. Since $x_n = n$, the sequence consists of distinct natural numbers. Consequently only finitely many terms of the sequence can belong to the finite set $\mathbb{R} \setminus U$. Therefore, $\exists\, N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow n \notin \mathbb{R} \setminus U.$$

Thus, $n \geq N \Rightarrow x_n = n \in U$. So, by the definition of convergence,

$$x_n \to x \quad \forall\, x \in \mathbb{R}.$$

Now, the sequence $x_n = n$ converges to every real number

$\Rightarrow$ its limit is not unique

$\Rightarrow$ the space cannot be Hausdorff.

Thus $(\mathbb{R}, \tau)$ is $T_1$ but not $T_2$, i.e. $T_1 \not\Rightarrow T_2$.

## Problem 3.2

Prove that the product topology $\mathcal{T}_{\text{prod}}$ on $\mathbb{R}^n$ (where each factor $\mathbb{R}$ is equipped with its standard Euclidean topology) coincides with the Euclidean topology $\mathcal{T}_{\text{Euc}}$ on $\mathbb{R}^n$ induced by the standard Euclidean metric.

### Solution

*submitted by* **Arunodaya**

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Let $U = U_1 \times U_2 \times \cdots \times U_n$ be a basic open set in the product topology. Take an arbitrary point $x = (x_1, x_2, x_3, x_4, \ldots, x_n) \in U$. Since each $U_i$ is open in $\mathbb{R}$, $\exists\, \varepsilon_i > 0$ such that

$$(x_i - \varepsilon_i,\, x_i + \varepsilon_i) \subseteq U_i.$$

Choose $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n\}$. Since there are finitely many positive numbers, $\varepsilon > 0$.

Now, consider any point $y = (y_1, y_2, \ldots, y_n) \in B(x, \varepsilon)$. Then $d(x, y) < \varepsilon$. Thus,

$$\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} < \varepsilon \;\Rightarrow\; \sum_{i=1}^{n}(x_i - y_i)^2 < \varepsilon^2.$$

Since every term in the sum is non-negative, $(x_i - y_i)^2 < \varepsilon^2$ for every $i = 1, 2, \ldots, n$. Consequently $|x_i - y_i| < \varepsilon \leq \varepsilon_i$; thus $y_i \in (x_i - \varepsilon_i,\, x_i + \varepsilon_i) \subseteq U_i$. This holds for every $i$. Hence

$$y \in U_1 \times U_2 \times \cdots \times U_n \;\Rightarrow\; B(x, \varepsilon) \subseteq U.$$

Since $x \in U$ was arbitrary, every point of $U$ has a Euclidean open ball contained in $U$. Hence $U$ is Euclidean open.

$$\underline{\mathcal{T}_{\text{prod}} \subseteq \mathcal{T}_{\text{Euc}}}$$

Let $V$ be an arbitrary Euclidean open subset of $\mathbb{R}^n$. Take any point $x = (x_1, x_2, \ldots, x_n) \in V$. Since $V$ is Euclidean open, $\exists\, \delta > 0$ such that $B(x, \delta) \subseteq V$. Now consider the product of open intervals

$$W = \prod_{i=1}^{n}\Big(x_i - \frac{\delta}{\sqrt{n}},\; x_i + \frac{\delta}{\sqrt{n}}\Big).$$

We will show that $W \subseteq B(x, \delta)$. Take any $y = (y_1, y_2, y_3, y_4, \ldots, y_n) \in W$. By definition of $W$,

$$|x_i - y_i| < \delta/\sqrt{n} \quad \forall\, i = 1, 2, 3, 4, \ldots, n.$$

Therefore

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} < \sqrt{\sum_{i=1}^{n}\frac{\delta^2}{n}} = \sqrt{\frac{n\delta^2}{n}} = \delta.$$

Thus $y \in B(x, \delta)$. Consequently $W \subseteq B(x, \delta) \subseteq V$.

Notice that $W$ is a basic open set in the product topology because it is a product of open intervals in $\mathbb{R}$. Hence, every point $x \in V$ has a basic product open neighbourhood contained in $V$. Therefore, $V$ is open in the product topology. So, $\underline{\mathcal{T}_{\text{Euc}} \subseteq \mathcal{T}_{\text{prod}}}$.

So $\mathcal{T}_{\text{prod}} = \mathcal{T}_{\text{Euc}}$.

## Problem 3.3

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and let $\phi\colon \Lambda \to \Lambda$ be a bijection. Prove that the product spaces $X = \prod_{\alpha\in\Lambda} X_\alpha$ and $X_\phi = \prod_{\alpha\in\Lambda} X_{\phi(\alpha)}$, both equipped with their respective product topologies, are homeomorphic.

### Solution

*submitted by* **Arunodaya**

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We define $f\colon X \to X_\phi$ by

$$f\big((x_\alpha)_{\alpha\in A}\big) = \big(x_{\phi(\alpha)}\big)_{\alpha\in A}.$$

Since $\phi$ is a bijection, every coordinate in the second product corresponds to exactly one coordinate in the first product. Since $\phi$ is bijective, it has an inverse

$$\phi^{-1}\colon B \to A.$$

Define $g\colon X_\phi \to X$ by

$$g\big((y_\beta)_{\beta\in B}\big) = \big(y_{\phi^{-1}(\beta)}\big)_{\beta\in B}.$$

Then for every $x = (x_\alpha) \in X$, $g(f(x)) = x$. Similarly for every $y \in X_\phi$, $f(g(y)) = y$. Thus $g = f^{-1}$. Hence $f$ is bijective.

The product topology has a subbasis consisting of sets of the form $\pi_\alpha^{-1}(U_\alpha)$, where $U_\alpha$ is open in $X_\alpha$ and $\pi_\alpha\colon X \to X_\alpha$ is the coordinate projection.

Let $U_\alpha$ be open in $X_{\phi(\alpha)}$. Consider the subbasic open set in $X_\phi$

$$V = \{y \in X_\phi : y_\alpha \in U_\alpha\}.$$

$$\begin{aligned} f^{-1}(V) &= \{x \in X : f(x)_\alpha \in U_\alpha\} \\ &= \{x \in X : x_\alpha \in U_\alpha\} \\ &= \pi_\alpha^{-1}(U_\alpha). \end{aligned}$$

Since $U_\alpha$ is open and $\pi_\alpha$ is continuous, the set is open in $X \Rightarrow$ the inverse image of every subbasic open set is open. Consequently, $f$ is continuous.

By the same argument, for every subbasic open set

$$W = \{x \in X : x_\alpha \in U_\alpha\}$$

we have

$$(f^{-1})^{-1}(W) = \{y \in X_\phi : y_{\phi^{-1}(\alpha)} \in U_\alpha\};$$

this is a subbasic open set in $X_\phi$. Hence $f^{-1}$ is continuous.

We proved that $f$ is bijective, continuous and has a continuous inverse; therefore $X \cong X_\phi$.

## Problem 3.4

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and for each $\alpha \in \Lambda$, let $Y_\alpha \subseteq X_\alpha$ be equipped with the subspace topology. Prove that the product topology on $Y = \prod_{\alpha\in\Lambda} Y_\alpha$ coincides with the subspace topology on $Y$ induced by the product topology on $X = \prod_{\alpha\in\Lambda} X_\alpha$.

### Solution

*submitted by* **Arunodaya**

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Let

- $\tau_1$ denote the product topology on $Y$,
- $\tau_2$ denote the subspace topology on $Y$ inherited from $X$.

We shall prove $\tau_1 = \tau_2$.

Let $B$ be a basic open set in $X$. Then

$$B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha),$$

where each $U_\alpha$ is open in $X_\alpha$. By the definition of the subspace topology, a basic open set in $Y$ is $B \cap Y$. Now,

$$\begin{aligned} B \cap Y &= Y \cap \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha) \\ &= \bigcap_{\alpha\in F} \left(Y \cap \pi_\alpha^{-1}(U_\alpha)\right) \\ &= \bigcap_{\alpha\in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha). \end{aligned}$$

Since $U_\alpha$ is open in $X_\alpha$, the set $U_\alpha \cap Y_\alpha$ is open in $Y_\alpha$ by the subspace topology. Therefore, $B \cap Y$ is open in the product topology on $Y$. Hence $\underline{\tau_2 \subseteq \tau_1}$.

Let $C$ be a basic open set in the product topology on $Y$. Then

$$C = \bigcap_{\alpha\in F} q_\alpha^{-1}(V_\alpha),$$

where $V_\alpha$ is open in $Y_\alpha$. Since $Y_\alpha$ has the subspace topology inherited from $X_\alpha$, $\exists$ an open set $U_\alpha$ in $X_\alpha$ such that $V_\alpha = U_\alpha \cap Y_\alpha$. Therefore,

$$\begin{aligned} C &= \bigcap_{\alpha\in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha) \\ &= Y \cap \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha). \end{aligned}$$

But $\bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha)$ is open in $X$ because it is a basic open set in the product topology on $X$. Thus, $C$ is the intersection of $Y$ with an open set in $X$. Hence $C$ is open in the subspace topology on $Y$. Therefore $\underline{\tau_1 \subseteq \tau_2}$.

So, $\underline{\tau_1 = \tau_2}$.

## Problem 3.5

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be an indexed collection of topological spaces, and let $X = \prod_{\alpha\in\Lambda} X_\alpha$ be equipped with the product topology $\mathcal{T}_{\text{prod}}$. For any fixed index $\beta \in \Lambda$, consider the $\beta$-th coordinate projection map

$$p_\beta \colon \prod_{\alpha\in\Lambda} X_\alpha \longrightarrow X_\beta, \qquad p_\beta\big((x_\alpha)_{\alpha\in\Lambda}\big) = x_\beta.$$

Prove that $p_\beta$ is an open map. Is $p_\beta$ necessarily a closed map?

### Solution

*submitted by* **Arunodaya**

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**Part (i).** By the definition of the product topology, every basic open set has the form

$$B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha),$$

where

- $F \subseteq A$ is finite,
- each $U_\alpha$ is open in $X_\alpha$.

**Case 1:** $\beta \in F$. Then $B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha)$, and the image of $B$ under $p_\beta$ is

$$p_\beta(B) = U_\beta,$$

provided the other coordinate restrictions are non-empty. Indeed, every element of $B$ has its $\beta$-coordinate in $U_\beta$. Conversely, any element of $U_\beta$ can be extended to a point of $B$ if all the other coordinate restrictions are non-empty. Since $U_\beta$ is open in $X_\beta$, the image is open. If another coordinate restriction is empty, then $B = \emptyset$, whose image is also open.

**Case 2:** $\beta \notin F$. In this case, there is no restriction on the $\beta$-coordinate:

$$p_\beta(B) = X_\beta,$$

provided $B \neq \emptyset$. Since $X_\beta$ is open in itself, the image is open. If $B = \emptyset$, its image is open again.

Now, $U = \bigcup_{\lambda\in\Lambda} B_\lambda$. Using the fact that images preserve arbitrary unions, we get

$$p_\beta(U) = p_\beta\Big(\bigcup_{\lambda\in\Lambda} B_\lambda\Big) = \bigcup_{\lambda\in\Lambda} p_\beta(B_\lambda).$$

From the above step, $p_\beta(B_\lambda)$ is open in $X_\beta$. An arbitrary union of open sets is open; thus, $p_\beta(U)$ is open in $X_\beta$.

**Part (ii). No.** Take $X_1 = \mathbb{R}$, $X_2 = \mathbb{R}$ with their usual Euclidean topologies. Then $X = X_1 \times X_2 = \mathbb{R}^2$. Consider the coordinate projection $p_1 \colon \mathbb{R}^2 \to \mathbb{R}$ defined by $p_1(x, y) = x$. Consider

$$F = \{(x, y) \in \mathbb{R}^2 : xy = 1\}.$$

Define $h \colon \mathbb{R}^2 \to \mathbb{R}$ by $h(x, y) = xy$. Since multiplication is continuous, $h$ is continuous. Also $F = h^{-1}(\{1\})$. Since $\{1\}$ is closed in $\mathbb{R}$ and the inverse image of a closed set under a continuous function is closed, we say $F$ is closed in $\mathbb{R}^2$.

We have

$$p_1(F) = \{x \in \mathbb{R} \mid \exists\, y \in \mathbb{R} \text{ such that } xy = 1\} = \mathbb{R} \setminus \{0\}.$$

Indeed, if $x \neq 0$ we can choose $y = 1/x$. But if $x = 0$, no such $y$ exists. Therefore $p_1(F) = \mathbb{R} \setminus \{0\}$.

The set $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$ is not closed in $\mathbb{R}$ because its complement $\{0\}$ is not open in $\mathbb{R}$. Thus, we found a closed set $F \subseteq \mathbb{R}^2$ whose image under $p_1$ is not closed. Consequently, $p_1$ is not closed.

So, every coordinate projection is an open map. A coordinate projection need not be a closed map.

## Problem 3.6

Let $L$ be a line in the plane $\mathbb{R}^2$, and $\mathbb{R}_\ell$ be the lower limit topology on $\mathbb{R}$. Discuss the topology $L$ inherits as a subspace of $\mathbb{R}_\ell \times \mathbb{R}$ and as a subspace of $\mathbb{R}_\ell \times \mathbb{R}_\ell$.

### Solution

*submitted by* **Pula Karan**

We analyze the subspace topology on $L$ under both product topologies.

**Part 1: Subspace Topology inherited from $\mathbb{R}_l \times \mathbb{R}$.** A basis for $\mathbb{R}_l \times \mathbb{R}$ consists of rectangles of the form $[a, b) \times (c, d)$ with $a < b$ and $c < d$.

(a) **Vertical Line ($L : x = k$):** For any $(k, y_0) \in L$, a basic open neighborhood in $\mathbb{R}_l \times \mathbb{R}$ is $U = [k, k + \epsilon) \times (y_0 - \epsilon, y_0 + \epsilon)$. Intersecting $U$ with $L$ yields:

$$U \cap L = \{k\} \times (y_0 - \epsilon, y_0 + \epsilon)$$

This traces open intervals along the $y$-axis. Thus, $L$ inherits the **standard Euclidean topology** ($\mathbb{R}$).

(b) **Horizontal Line ($L : y = k$):** For any $(x_0, k) \in L$, a basic open neighborhood in $\mathbb{R}_l \times \mathbb{R}$ is $U = [x_0, x_0 + \epsilon) \times (k - \epsilon, k + \epsilon)$. Intersecting $U$ with $L$ yields:

$$U \cap L = [x_0, x_0 + \epsilon) \times \{k\}$$

This traces right-half-open intervals along the $x$-axis. Thus, $L$ inherits the **lower limit topology** ($\mathbb{R}_l$).

(c) **Non-Vertical Line ($L : y = mx + k$, $m \neq 0$):** For any $(x_0, y_0) \in L$, consider basic open sets $U = [x_0, x_0 + \epsilon) \times (y_0 - \delta, y_0 + \delta)$.

- If $m > 0$, as $x$ increases from $x_0$, $y$ increases from $y_0$. For sufficiently small $\delta > 0$, $U \cap L$ isolates points corresponding to $x \in [x_0, x_0 + \delta/m)$.
- If $m < 0$, as $x$ increases from $x_0$, $y$ decreases from $y_0$. Intersecting $[x_0, x_0 + \epsilon)$ with $(y_0 - \delta, y_0 + \delta)$ again isolates points corresponding to $x \in [x_0, x_0 + \delta/|m|)$.

In both cases, basic open sets induce half-open intervals $[x_0, x_0 + r)$ along $L$. Thus, $L$ inherits the **lower limit topology** ($\mathbb{R}_l$).

**Part 2: Subspace Topology inherited from $\mathbb{R}_l \times \mathbb{R}_l$ (Sorgenfrey Plane).** A basis for $\mathbb{R}_l \times \mathbb{R}_l$ consists of half-open rectangles of the form $[a, b) \times [c, d)$.

(a) **Vertical ($x = k$), Horizontal ($y = k$), and Positive Slope ($y = mx + k$, $m > 0$):** For any $(x_0, y_0) \in L$, intersecting $[x_0, x_0 + \epsilon) \times [y_0, y_0 + \epsilon')$ with $L$ isolates points corresponding to $[x_0, x_0 + \delta)$ where $\delta = \min(\epsilon, \epsilon'/m)$ for $m > 0$ (or $\epsilon$ if vertical/horizontal). Thus, $L$ inherits the **lower limit topology** ($\mathbb{R}_l$).

(b) **Oblique Line with Negative Slope ($y = mx + k$, $m < 0$):** Let $(x_0, y_0) \in L$ and consider $U = [x_0, x_0 + \epsilon) \times [y_0, y_0 + \epsilon)$. For any $x > x_0$, $y = mx + k < mx_0 + k = y_0$, so $(x, y) \notin U$. For $x < x_0$, $x \notin [x_0, x_0 + \epsilon)$, so $(x, y) \notin U$. Therefore, $U \cap L = \{(x_0, y_0)\}$. Since every singleton is open, $L$ inherits the **discrete topology**.

## Problem 3.7

Compare (i) the ordered square $I_o^2$, (ii) the topology $I^2$ inherits as a subspace of the product topology $\mathbb{R}^2$, and (iii) the product topology on $I \times I$.

### Solution

*submitted by* **Pula Karan**

Let $I = [0, 1]$. We compare the three topologies on $I^2 = [0, 1] \times [0, 1]$:

1. **Topologies (ii) and (iii) are identical ($\mathcal{T}_{\text{sub}} = \mathcal{T}_{\text{prod}}$):**
   - Topology (ii) is the subspace topology inherited from the Euclidean product topology on $\mathbb{R}^2$. Basis elements are of the form $(U \times V) \cap I^2$, where $U, V \subseteq \mathbb{R}$ are open intervals.
   - Topology (iii) is the product topology on $I \times I$, where each factor $I$ carries the subspace topology from $\mathbb{R}$. Basis elements are of the form $(U \cap I) \times (V \cap I) = (U \times V) \cap I^2$.

   Since their bases coincide, Topologies (ii) and (iii) are identical.
2. **Topology (i) ($I_o^2$) is strictly finer than Topologies (ii) and (iii):** The ordered square $I_o^2$ is equipped with the dictionary order topology $\leq_{\text{dic}}$.
   - *Euclidean $\subseteq$ Dictionary Order*: Any standard Euclidean basic open set $U = (a, b) \times (c, d) \subset (0, 1)^2$ can be written as:

     $$U = \bigcup_{x_0 \in (a,b)} \Big( \{x_0\} \times (c, d) \Big) = \bigcup_{x_0 \in (a,b)} \Big( (x_0, c), (x_0, d) \Big)_{\text{dic}}$$

     Since each vertical interval $\{x_0\} \times (c, d)$ is open in the dictionary order topology, $U$ is open in $I_o^2$.
   - *Strict Containment*: Sets of the form $\{x_0\} \times (c, d)$ for $0 < c < d < 1$ are open in $I_o^2$. However, they contain no 2D Euclidean open ball, so they are not open in the standard topology on $I^2$.

**Conclusion:** Topology (ii) = Topology (iii) $\subsetneq$ Topology (i).

## Problem 3.8

Let $X$ be a topological space, and $\Delta = \{(x, x) \in X \times X \mid x \in X\}$ be the diagonal of $X \times X$, equipped with the subspace topology inherited from the product topology on $X \times X$.

(i) Prove that $X$ is Hausdorff if and only if the diagonal $\Delta$ is a closed subset of $X \times X$.

(ii) Prove that the diagonal map $d\colon X \to \Delta$ defined by $d(x) = (x, x)$ is a homeomorphism.

**Solution** *submitted by* **Pula Karan**

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### Part (i).

*Proof.* ( $\implies$ ) Suppose $X$ is Hausdorff. We show that $(X \times X) \setminus \Delta$ is open. Let $(x, y) \in (X \times X) \setminus \Delta$, which means $x \neq y$. Since $X$ is Hausdorff, there exist disjoint open sets $U, V \subseteq X$ such that $x \in U$, $y \in V$, and $U \cap V = \varnothing$. Then $U \times V$ is an open neighborhood of $(x, y)$ in $X \times X$. If there were some $(z, z) \in (U \times V) \cap \Delta$, then $z \in U \cap V = \varnothing$, a contradiction. Hence $(U \times V) \cap \Delta = \varnothing$, so $U \times V \subseteq (X \times X) \setminus \Delta$. Thus $(X \times X) \setminus \Delta$ is open, so $\Delta$ is closed.

( $\impliedby$ ) Suppose $\Delta$ is closed in $X \times X$, so $(X \times X) \setminus \Delta$ is open. Let $x, y \in X$ with $x \neq y$. Then $(x, y) \in (X \times X) \setminus \Delta$. By definition of the product topology, there exists a basic open set $U \times V \subseteq X \times X$ such that $(x, y) \in U \times V \subseteq (X \times X) \setminus \Delta$. This implies $x \in U$ and $y \in V$. If there existed $z \in U \cap V$, then $(z, z) \in (U \times V) \cap \Delta$, contradicting $U \times V \subseteq (X \times X) \setminus \Delta$. Therefore $U \cap V = \varnothing$, so $X$ is Hausdorff. □

### Part (ii).

*Proof.* We check the three properties of a homeomorphism for $d\colon X \to \Delta$:

1. **Bijectivity:**
   - *Injective*: $d(x_1) = d(x_2) \implies (x_1, x_1) = (x_2, x_2) \implies x_1 = x_2$.
   - *Surjective*: Any element in $\Delta$ is of the form $(x, x) = d(x)$ for some $x \in X$.
2. **Continuity of $d$:** The component functions of $d$ are $d_1 = \mathrm{id}_X$ and $d_2 = \mathrm{id}_X$. Since the identity map $\mathrm{id}_X\colon X \to X$ is continuous, $d\colon X \to X \times X$ is continuous by the characteristic property of product spaces. Restricting the codomain to the subspace $\Delta$ preserves continuity.
3. **Continuity of $d^{-1}$:** The inverse map $d^{-1}\colon \Delta \to X$ is given by $d^{-1}(x, x) = x$. This is the restriction of the first coordinate projection $p_1\colon X \times X \to X$ to $\Delta$. Since projection maps are continuous, its restriction $d^{-1} = p_1|_\Delta$ is continuous.

Thus, $d\colon X \to \Delta$ is a homeomorphism. □

## Problem 3.9

Let $X$ and $Y$ be topological spaces, and let $f, g\colon X \to Y$ be continuous maps. Assume that $Y$ is a Hausdorff space. Prove the following statements:

(i) The coincidence set $E = \{x \in X \mid f(x) = g(x)\}$ is closed in $X$.

(ii) If $D$ is a dense subset of $X$ and $f|_D = g|_D$, then $f = g$ on all of $X$.

(iii) The graph of $f$, defined as $\Gamma_f = \{(x, f(x)) \in X \times Y \mid x \in X\}$, is closed in the product space $X \times Y$.

(iv) Assume that $f\colon X \to Y$ is a continuous, open, and surjective map. Then the fiber relation set $R = \{(x_1, x_2) \in X \times X \mid f(x_1) = f(x_2)\}$ is closed in $X \times X$ if and only if $Y$ is Hausdorff.

**Solution** *submitted by* **Pula Karan**

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### Part (i).

*Proof.* Define $h\colon X \to Y \times Y$ by $h(x) = (f(x), g(x))$. Since $f$ and $g$ are continuous, $h$ is continuous. Because $Y$ is Hausdorff, the diagonal $\Delta_Y = \{(y, y) \in Y \times Y\}$ is closed in $Y \times Y$ by Problem 3.8(i). Notice that:

$$x \in E \iff f(x) = g(x) \iff (f(x), g(x)) \in \Delta_Y \iff x \in h^{-1}(\Delta_Y)$$

Thus $E = h^{-1}(\Delta_Y)$. Since $\Delta_Y$ is closed and $h$ is continuous, $E$ is closed in $X$. □

### Part (ii).

*Proof.* Consider the coincidence set $E = \{x \in X \mid f(x) = g(x)\}$. By part (i), $E$ is closed in $X$. Since $f|_D = g|_D$, we have $D \subseteq E$. Taking closures gives $\overline{D} \subseteq \overline{E} = E$. Since $D$ is dense in $X$, $\overline{D} = X$. Therefore $X \subseteq E$, which means $E = X$, so $f(x) = g(x)$ for all $x \in X$. □

### Part (iii).

*Proof.* Define $F\colon X \times Y \to Y \times Y$ by $F(x, y) = (f(x), y)$. $F$ is continuous because its component maps $f \circ p_1$ and $p_2$ are continuous. Since $Y$ is Hausdorff, $\Delta_Y$ is closed in $Y \times Y$. Notice that:

$$(x, y) \in \Gamma_f \iff y = f(x) \iff (f(x), y) \in \Delta_Y \iff F(x, y) \in \Delta_Y$$

Thus $\Gamma_f = F^{-1}(\Delta_Y)$. Since $F$ is continuous and $\Delta_Y$ is closed, $\Gamma_f$ is closed in $X \times Y$. □

### Part (iv).

*Proof.* Consider the product map $f \times f\colon X \times X \to Y \times Y$ defined by $(f \times f)(x_1, x_2) = (f(x_1), f(x_2))$. Since $f$ is continuous, open, and surjective, $f \times f$ is also continuous, open, and surjective. Notice that $R = (f \times f)^{-1}(\Delta_Y)$.

($\Longleftarrow$) If $Y$ is Hausdorff, $\Delta_Y$ is closed in $Y \times Y$. By continuity of $f \times f$, $R = (f \times f)^{-1}(\Delta_Y)$ is closed in $X \times X$.

( $\implies$ ) Suppose $R$ is closed in $X \times X$. Since $f \times f$ is surjective, $(f \times f)(R) = \Delta_Y$. Taking complements yields:

$$(Y \times Y) \setminus \Delta_Y = (f \times f)((X \times X) \setminus R)$$

Since $R$ is closed, $(X \times X) \setminus R$ is open. Because $f \times f$ is an open map, $(Y \times Y) \setminus \Delta_Y$ is open in $Y \times Y$, which means $\Delta_Y$ is closed in $Y \times Y$. By Problem 3.8(i), $Y$ is Hausdorff. □

## Problem 3.10

Let $X$ be a topological space and $A \subset X$ be dense in $X$. Let $Y$ be a Hausdorff space and $f\colon A \to Y$ be a continuous map. Show that if $f$ may be extended to a continuous map $g\colon X \to Y$, then $g$ is uniquely determined by $f$.

### Solution

*submitted by* **Pula Karan**

*Proof.* Suppose $g_1, g_2\colon X \to Y$ are two continuous extensions of $f\colon A \to Y$, so $g_1|_A = f$ and $g_2|_A = f$.

1. Since $g_1|_A = g_2|_A = f$, the maps $g_1$ and $g_2$ agree on the subset $A \subseteq X$.
2. $A$ is dense in $X$, and $Y$ is a Hausdorff space.
3. By **Problem 3.9(ii)**, if two continuous functions from $X$ into a Hausdorff space $Y$ agree on a dense subset $A \subseteq X$, they must agree on all of $X$.

Therefore, $g_1(x) = g_2(x)$ for all $x \in X$, proving that the continuous extension $g$ is uniquely determined by $f$. □

# Problem Set 4

## Problem 4.1

Let $X$ and $Y$ be two topological spaces; and $a \in X$, $b \in Y$ be two given points. Prove that the maps $f\colon X \to X \times Y$ and $g\colon Y \to X \times Y$ given by

$$f(x) = (x, b) \quad \text{and} \quad g(y) = (a, y)$$

are embeddings.

### Solution

*submitted by* **Dhrubojoti**

**Proof.** Recall that a map $h\colon Z \to W$ is an embedding if it is a homeomorphism from $Z$ onto $h(Z)$, equipped with the subspace topology.

**1. The map $f\colon X \to X \times Y$, $f(x) = (x, b)$.** *Step 1: $f$ is continuous.* Let $U \times V$ be an arbitrary basic open set in $X \times Y$, where $U \subseteq X$ and $V \subseteq Y$ are open. Then

$$\begin{aligned} f^{-1}(U \times V) &= \{x \in X : f(x) \in U \times V\} \\ &= \{x \in X : (x, b) \in U \times V\} \\ &= \begin{cases} U, & b \in V, \\ \emptyset, & b \notin V. \end{cases} \end{aligned}$$

In either case, $f^{-1}(U \times V)$ is open in $X$. Hence $f$ is continuous.

*Step 2: $f$ is injective.* Suppose that $f(x_1) = f(x_2)$. Then

$$(x_1, b) = (x_2, b).$$

Therefore, $x_1 = x_2$, so $f$ is injective.

*Step 3: The inverse map $f^{-1}\colon f(X) \to X$ is continuous.* Observe that

$$f(X) = X \times \{b\}.$$

Consider the first projection map

$$\pi_X\colon X \times Y \to X, \qquad \pi_X(x, y) = x.$$

Since projection maps are continuous, its restriction

$$\pi_X|_{f(X)}\colon f(X) \to X$$

is continuous. But

$$\pi_X(f(x)) = \pi_X(x, b) = x.$$

Thus

$$f^{-1} = \pi_X|_{f(X)}.$$

Hence $f^{-1}$ is continuous.

Therefore, $f$ is a homeomorphism from $X$ onto $f(X)$, and consequently $f$ is an embedding.

**2. The map $g\colon Y \to X \times Y$, $g(y) = (a, y)$.** *Step 1: $g$ is continuous.* Let $U \times V$ be an arbitrary basic open set in $X \times Y$. Then

$$\begin{aligned} g^{-1}(U \times V) &= \{y \in Y : g(y) \in U \times V\} \\ &= \{y \in Y : (a, y) \in U \times V\} \\ &= \begin{cases} V, & a \in U, \\ \emptyset, & a \notin U. \end{cases} \end{aligned}$$

In either case, $g^{-1}(U \times V)$ is open in $Y$. Hence $g$ is continuous.

*Step 2: $g$ is injective.* Suppose that $g(y_1) = g(y_2)$. Then

$$(a, y_1) = (a, y_2).$$

Therefore, $y_1 = y_2$, so $g$ is injective.

*Step 3: The inverse map $g^{-1}\colon g(Y) \to Y$ is continuous.* Observe that

$$g(Y) = \{a\} \times Y.$$

Consider the second projection map

$$\pi_Y\colon X \times Y \to Y, \qquad \pi_Y(x, y) = y.$$

Since projection maps are continuous, its restriction

$$\pi_Y|_{g(Y)}\colon g(Y) \to Y$$

is continuous. But

$$\pi_Y(g(y)) = \pi_Y(a, y) = y.$$

Thus

$$g^{-1} = \pi_Y|_{g(Y)}.$$

Hence $g^{-1}$ is continuous.

Therefore, $g$ is a homeomorphism from $Y$ onto $g(Y)$, and consequently $g$ is an embedding.

Hence both $f$ and $g$ are embeddings.

## Problem 4.2

Suppose that $f, g\colon X \to \mathbb{R}$ are two continuous maps.

(i). Prove that the set $\{x \in X \mid f(x) \leq g(x)\}$ is closed in $X$.

(ii). Prove that the map $h\colon X \to \mathbb{R}$ given by $h(x) = \min\{f(x), g(x)\}$ is continuous.

**Solution** *submitted by* **Dhrubojoti**

**Proof.**

**(i) The set $\{x \in X \mid f(x) \le g(x)\}$ is closed in $X$.** Let

$$A = \{x \in X \mid f(x) \le g(x)\}.$$

We know that $f$ and $g$ are continuous. Since the subtraction map

$$\varphi\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}, \qquad \varphi(u, v) = u - v$$

is continuous, the map

$$F\colon X \to \mathbb{R}, \qquad F(x) = f(x) - g(x)$$

is continuous. Now observe that

$$\begin{aligned} A &= \{x \in X \mid f(x) \le g(x)\} \\ &= \{x \in X \mid f(x) - g(x) \le 0\} \\ &= F^{-1}((-\infty, 0]). \end{aligned}$$

Since $(-\infty, 0]$ is closed in $\mathbb{R}$ and $F$ is continuous, its inverse image is closed in $X$. Therefore,

$$\boxed{\{x \in X \mid f(x) \le g(x)\} \text{ is closed in } X.}$$

**(ii) The map $h\colon X \to \mathbb{R}$ given by $h(x) = \min\{f(x), g(x)\}$ is continuous.** We use the identity

$$\min\{a, b\} = \frac{a + b - |a - b|}{2}, \qquad a, b \in \mathbb{R}.$$

Therefore, for every $x \in X$,

$$h(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}.$$

Since $f$ and $g$ are continuous, the maps

$$f + g\colon X \to \mathbb{R} \quad \text{and} \quad f - g\colon X \to \mathbb{R}$$

are continuous. Also, the absolute value map

$$|\cdot|\colon \mathbb{R} \to \mathbb{R}, \qquad t \mapsto |t|$$

is continuous. Hence the composition

$$x \mapsto |f(x) - g(x)|$$

is continuous. Consequently,

$$h(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}$$

is continuous, since sums, differences and scalar multiples of continuous real-valued functions are continuous. Thus,

$$\boxed{h\colon X \to \mathbb{R} \text{ is continuous.}}$$

## Problem 4.3

Let $\{A_\alpha\}$ be a collection of subsets of $X$ such that $X = \bigcup_\alpha A_\alpha$. Suppose that $f\colon X \to Y$ is a map such that the restriction map $f|_{A_\alpha}\colon A_\alpha \to Y$ is continuous for each $\alpha$.

(i). Prove that if the collection $\{A_\alpha\}$ is finite and each set $A_\alpha$ is closed, then $f$ is continuous.

(ii). Find an example where the collection $\{A_\alpha\}$ is countable and each set $A_\alpha$ is closed, but $f$ is not continuous.

(iii). An indexed family of sets $\{A_\alpha\}$ is said to be *locally finite* if each point $x$ has a neighborhood that intersects $A_\alpha$ for only finitely many values of $\alpha$. Prove that if the family $\{A_\alpha\}$ is locally finite and each set $A_\alpha$ is closed, then $f$ is continuous.

### Solution

*submitted by* **Dhrubojoti**

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**Recall.** A map $f\colon X \to Y$ is continuous if and only if the preimage of every closed subset of $Y$ is closed in $X$. Also, if $A$ is a closed subset of $X$, then every relatively closed subset of $A$ is closed in $X$.

#### (i) The collection $\{A_\alpha\}$ is finite and each $A_\alpha$ is closed.

*Proof.* Suppose

$$X = A_1 \cup A_2 \cup \cdots \cup A_n,$$

where each $A_i$ is closed in $X$. We are given that the restriction

$$f|_{A_i}\colon A_i \to Y$$

is continuous for every $i = 1, 2, \ldots, n$. We need to prove that $f\colon X \to Y$ is continuous.

Let $C$ be an arbitrary closed subset of $Y$. Consider

$$f^{-1}(C) = \{x \in X : f(x) \in C\}.$$

Since $X = \bigcup_{i=1}^n A_i$, we can write

$$f^{-1}(C) = \bigcup_{i=1}^{n} (f^{-1}(C) \cap A_i).$$

Now observe that

$$f^{-1}(C) \cap A_i = (f|_{A_i})^{-1}(C).$$

Since $f|_{A_i}$ is continuous and $C$ is closed in $Y$, $(f|_{A_i})^{-1}(C)$ is closed in $A_i$. But $A_i$ itself is closed in $X$. Therefore, a relatively closed subset of $A_i$ is also closed in $X$. Thus, $f^{-1}(C) \cap A_i$ is closed in $X$ for every $i$. Consequently,

$$f^{-1}(C) = \bigcup_{i=1}^{n} (f^{-1}(C) \cap A_i)$$

is a finite union of closed subsets of $X$. Hence, $f^{-1}(C)$ is closed in $X$. Since $C$ was an arbitrary closed subset of $Y$, we conclude that

$$\boxed{f\colon X \to Y \text{ is continuous.}}$$

□

**(ii) Find a countable collection of closed sets covering $X$ for which $f$ is not continuous. Example.** Take

$$X = \mathbb{R}, \qquad Y = \mathbb{R}.$$

Define

$$A_0 = \{0\}$$

and, for $n \in \mathbb{N}$,

$$A_n = \left[-n, -\frac{1}{n}\right] \cup \left[\frac{1}{n}, n\right].$$

**Step 1. Show that each $A_n$ is closed.** Clearly, $A_0 = \{0\}$ is closed in $\mathbb{R}$. Also, each $A_n$ is a finite union of closed intervals. Hence, every $A_n$ is closed in $\mathbb{R}$.

**Step 2. Show that the collection is countable and covers $\mathbb{R}$.** The collection

$$\{A_0, A_1, A_2, \dots\}$$

is countable. Moreover, every nonzero real number belongs to some $A_n$. Indeed, given $x \neq 0$, choose $n$ sufficiently large such that

$$n \geq |x| \quad \text{and} \quad \frac{1}{n} \leq |x|.$$

Then $x \in A_n$. Since $0 \in A_0$, we obtain

$$\boxed{\mathbb{R} = \bigcup_{n=0}^{\infty} A_n.}$$

**Step 3. Define the map $f$.** Define $f\colon \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

We now show that the restriction of $f$ to every $A_n$ is continuous.

- On $A_0 = \{0\}$, the restriction is the constant function $f(0) = 1$, which is continuous.
- On every $A_n$, $n \geq 1$, we have $0 \notin A_n$. Therefore, $f|_{A_n}$ is the constant zero function, which is continuous.

Thus, every restriction $f|_{A_n}\colon A_n \to \mathbb{R}$ is continuous.

**Step 4. Show that $f$ is not continuous.** Consider the open set

$$U = \left(\frac{1}{2}, \frac{3}{2}\right) \subseteq \mathbb{R}.$$

We have

$$f^{-1}(U) = \{0\}.$$

But $\{0\}$ is not open in $\mathbb{R}$. Therefore, $f$ is not continuous.

Hence, we have constructed a countable closed cover of $\mathbb{R}$ such that every restriction is continuous, but the original map is not continuous. Thus,

$$\boxed{\text{The conclusion of (i) need not hold for a countable cover.}}$$

□

**(iii) Prove the result when $\{A_\alpha\}$ is locally finite and each $A_\alpha$ is closed.**

*Proof.* We are given that:

1. $X = \bigcup_\alpha A_\alpha$.
2. Every $A_\alpha$ is closed in $X$.
3. The family $\{A_\alpha\}$ is locally finite.
4. Every restriction $f|_{A_\alpha} : A_\alpha \to Y$ is continuous.

We need to prove that $f : X \to Y$ is continuous. We will use the following important result.

**Lemma.** *The union of a locally finite family of closed subsets of a topological space is closed.*

*Proof of the lemma.* Let $\{F_\alpha\}$ be a locally finite family of closed subsets of $X$. We want to show that

$$F = \bigcup_\alpha F_\alpha$$

is closed in $X$. Take an arbitrary point $x \in X \setminus F$. Since the family is locally finite, there exists an open neighbourhood $U$ of $x$ that intersects only finitely many members of the family. Suppose these members are

$$F_{\alpha_1}, F_{\alpha_2}, \ldots, F_{\alpha_n}.$$

Now, $x \notin F$, so in particular,

$$x \notin F_{\alpha_i}, \qquad i = 1, \ldots, n.$$

Since each $F_{\alpha_i}$ is closed, its complement is open. Thus, $X \setminus F_{\alpha_i}$ is an open neighbourhood of $x$. Consider

$$V = U \cap \bigcap_{i=1}^{n} (X \setminus F_{\alpha_i}).$$

This is an open neighbourhood of $x$, being a finite intersection of open sets. We claim that

$$V \cap F = \emptyset.$$

Indeed:

- For the finitely many sets $F_{\alpha_1}, \ldots, F_{\alpha_n}$, the set $V$ does not intersect any of them.
- For every other index $\alpha$, $U$, and hence $V$, does not intersect $F_\alpha$, by the choice of $U$.

Therefore, $V$ does not intersect their union $F$. So every point in $X \setminus F$ has an open neighbourhood contained in $X \setminus F$. Hence, $X \setminus F$ is open, which means

$$\bigcup_\alpha F_\alpha \text{ is closed in } X.$$

This proves the lemma.

Now we prove the required result. Let $C$ be an arbitrary closed subset of $Y$. Consider $f^{-1}(C)$. Since $X = \bigcup_\alpha A_\alpha$, we have

$$\begin{aligned} f^{-1}(C) &= \bigcup_\alpha (f^{-1}(C) \cap A_\alpha) \\ &= \bigcup_\alpha (f|_{A_\alpha})^{-1}(C). \end{aligned}$$

Now, since every restriction $f|_{A_\alpha}$ is continuous and $C$ is closed in $Y$, $(f|_{A_\alpha})^{-1}(C)$ is closed in $A_\alpha$. Since $A_\alpha$ is closed in $X$, it follows that

$$F_\alpha := (f|_{A_\alpha})^{-1}(C)$$

is closed in $X$. Furthermore, since $F_\alpha \subseteq A_\alpha$ and the family $\{A_\alpha\}$ is locally finite, the family $\{F_\alpha\}$ is also locally finite. Therefore, by our lemma, the union

$$f^{-1}(C) = \bigcup_\alpha F_\alpha$$

is closed in $X$. Since $C$ was an arbitrary closed subset of $Y$, the preimage of every closed subset of $Y$ is closed in $X$. Consequently,

$$\boxed{f\colon X \to Y \text{ is continuous.}}$$ □

## Problem 4.4

Let $f\colon X \to A$ and $g\colon Y \to B$ be two continuous maps. Define $f \times g\colon X \times Y \to A \times B$ by $(f \times g)(x, y) = (f(x), g(y))$. Prove that $f \times g$ is a continuous map.

### Solution

*submitted by* **Dhrubojoti**

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*Proof.* We are given that

$$f\colon X \to A \quad \text{and} \quad g\colon Y \to B$$

are continuous maps. We need to prove that

$$f \times g\colon X \times Y \to A \times B$$

is continuous. Recall that the product topology on $A \times B$ has a basis consisting of sets of the form $U \times V$, where $U$ is open in $A$ and $V$ is open in $B$. We will use the characterization of continuity in terms of the preimages of basic open sets.

**Step 1. Consider an arbitrary basic open set in $A \times B$.** Let $U \times V$ be an arbitrary basic open set in $A \times B$, where $U \subseteq A$ and $V \subseteq B$ are open. We have

$$\begin{aligned}(f \times g)^{-1}(U \times V) &= \{(x, y) \in X \times Y : (f \times g)(x, y) \in U \times V\} \\ &= \{(x, y) \in X \times Y : (f(x), g(y)) \in U \times V\} \\ &= \{(x, y) \in X \times Y : f(x) \in U,\ g(y) \in V\} \\ &= f^{-1}(U) \times g^{-1}(V).\end{aligned}$$

Therefore,

$$(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V).$$

**Step 2. Show that the preimage is open in $X \times Y$.** Since $f\colon X \to A$ is continuous and $U$ is open in $A$, we know that $f^{-1}(U)$ is open in $X$. Similarly, since $g\colon Y \to B$ is continuous and $V$ is open in $B$, $g^{-1}(V)$ is open in $Y$. Hence, $f^{-1}(U) \times g^{-1}(V)$ is open in $X \times Y$, by the definition of the product topology. Consequently, $(f \times g)^{-1}(U \times V)$ is open in $X \times Y$.

**Step 3. Conclude continuity.** We have shown that the preimage under $f \times g$ of every basic open set in $A \times B$ is open in $X \times Y$. Since every open subset of $A \times B$ is a union of basic open sets, the preimage of every open subset of $A \times B$ is open in $X \times Y$. Therefore,

$$\boxed{f \times g\colon X \times Y \to A \times B \text{ is continuous.}}$$ □

## Problem 4.5

Let $(X, d)$ be a metric space and let $S \subseteq X$ be a non-empty subset. For any $x \in X$, define the distance from $x$ to $S$ by $d(x, S) = \inf_{s \in S} d(x, s)$.

(1) Show that for any $x, y \in X$, $|d(x, S) - d(y, S)| \leq d(x, y)$.

(2) Conclude that $f_S \colon X \to \mathbb{R}$ defined by $f_S(x) = d(x, S)$ is continuous on $X$.

### Solution

*submitted by* **Aaditya**

**Proof.**

(1) By the triangle inequality, for every $s \in S$,

$$d(x, s) \leq d(x, y) + d(y, s).$$

Taking the infimum over $s \in S$, we get

$$d(x, S) \leq d(x, y) + d(y, S).$$

Thus,

$$d(x, S) - d(y, S) \leq d(x, y).$$

Interchanging $x$ and $y$ gives

$$d(y, S) - d(x, S) \leq d(x, y).$$

Therefore,

$$\boxed{|d(x, S) - d(y, S)| \leq d(x, y).}$$

(2) Let $x_0 \in X$ and $\varepsilon > 0$. Choose $\delta = \varepsilon$.

If $d(x, x_0) < \delta$, then by (1),

$$\begin{aligned} |f_S(x) - f_S(x_0)| &= |d(x, S) - d(x_0, S)| \\ &\leq d(x, x_0) \\ &< \varepsilon. \end{aligned}$$

Hence, $f_S$ is continuous on $X$. □

## Problem 4.6

Let $\{A_n\}_{n\in\mathbb{N}}$ be a sequence of connected subspaces of $X$ such that $A_n \cap A_{n+1} \neq \emptyset$ for all $n \in \mathbb{N}$. Prove that $\bigcup_{n=1}^{\infty} A_n$ is connected.

### Solution

*submitted by* **Aaditya**

**Proof.**

Let

$$A = \bigcup_{n=1}^{\infty} A_n.$$

Suppose, for contradiction, that $A$ is disconnected. Then

$$A = U \cup V,$$

where $U, V$ are non-empty, disjoint and open in $A$.

Since each $A_n$ is connected and

$$A_n = (A_n \cap U) \cup (A_n \cap V),$$

we must have

$$A_n \subseteq U \quad \text{or} \quad A_n \subseteq V.$$

Without loss of generality, let $A_1 \subseteq U$.

Since $A_1 \cap A_2 \neq \emptyset$, there exists $p \in A_1 \cap A_2$. Thus $p \in U$, so $A_2 \not\subseteq V$. Consequently, $A_2 \subseteq U$.

Repeating this argument inductively, we obtain

$$A_n \subseteq U \quad \text{for every } n \in \mathbb{N}.$$

Therefore,

$$A = \bigcup_{n=1}^{\infty} A_n \subseteq U,$$

contradicting $V \neq \emptyset$.

Hence,

$$\boxed{\bigcup_{n=1}^{\infty} A_n \text{ is connected.}}$$

□

## Problem 4.7

Determine whether or not $\mathbb{R}^\omega$ is connected under the uniform topology.

### Solution

*submitted by* **Aaditya**

**Proof.**

The uniform metric is

$$\bar{\rho}(x,y) = \sup_n \min\{|x_n - y_n|, 1\}.$$

Define

$$\mathcal{B} = \left\{ x \in \mathbb{R}^\omega : \sup_n |x_n| < \infty \right\}$$

and

$$\mathcal{U} = \mathbb{R}^\omega \setminus \mathcal{B}.$$

We show that both sets are open.

First, observe that

$$\bar{\rho}(x,y) < 1 \implies |x_n - y_n| < 1 \quad \forall n.$$

**Claim 1: $\mathcal{B}$ is open.**

Let $x \in \mathcal{B}$. Then $|x_n| \leq M$ for some $M > 0$ and all $n$.

If $\bar{\rho}(x,y) < 1$, then

$$|y_n| \leq |x_n| + |y_n - x_n| < M + 1.$$

Hence $y \in \mathcal{B}$, proving that $\mathcal{B}$ is open.

**Claim 2: $\mathcal{U}$ is open.**

Let $x \in \mathcal{U}$ and suppose $\bar{\rho}(x,y) < 1$.

If $y$ were bounded, say $|y_n| \leq M$, then

$$|x_n| \leq |y_n| + |x_n - y_n| < M + 1.$$

This contradicts the fact that $x$ is unbounded.

Thus $y \in \mathcal{U}$, proving that $\mathcal{U}$ is open.

Finally, $\mathcal{B}$ and $\mathcal{U}$ are disjoint, non-empty, and

$$\mathcal{B} \cup \mathcal{U} = \mathbb{R}^\omega.$$

Therefore, they form a separation of $\mathbb{R}^\omega$. Hence,

$$\boxed{\mathbb{R}^\omega \text{ is not connected.}}$$

□

## Problem 4.8

Determine whether or not the lower limit topology $\mathbb{R}_\ell$ on $\mathbb{R}$ is connected.

### Solution

*submitted by* **Aaditya**

**Proof.**

The lower limit topology has the basis

$$\mathcal{B} = \{[a, b) : a < b\}.$$

Consider

$$U = (-\infty, 0), \qquad V = [0, \infty).$$

We show that both sets are open.

If $x \in U$, then

$$x \in [x, 0) \subseteq U.$$

Since $[x, 0)$ is a basis element, $U$ is open.

Similarly, if $x \in V$, then

$$x \in [x, x + 1) \subseteq V.$$

Thus, $V$ is also open.

Moreover,

$$U \cap V = \emptyset, \qquad U \cup V = \mathbb{R}.$$

Both sets are non-empty. Therefore, $(U, V)$ forms a separation of $\mathbb{R}_\ell$.

Hence,

$$\boxed{\mathbb{R}_\ell \text{ is not connected.}}$$

□

## Problem 4.9

Let $C \subset X$ be a connected subspace of $X$ and $A \subset X$. Prove that if $C$ intersects both $A$ and $X \setminus A$, then $C$ intersects $\operatorname{Bd} A$.

### Solution

*submitted by* **Harisankar B**

Recall that $\operatorname{Bd} A = \overline{A} \cap \overline{X \setminus A}$, and that $X \setminus \overline{X \setminus A} = \operatorname{Int} A$ and $X \setminus \overline{A} = \operatorname{Int}(X \setminus A)$.

*Proof.* Suppose, for contradiction, that $C \cap \operatorname{Bd} A = \emptyset$. Consider the two sets

$$P = C \cap \operatorname{Int} A, \qquad Q = C \cap (X \setminus \overline{A}).$$

Both $\operatorname{Int} A$ and $X \setminus \overline{A}$ are open in $X$, so $P$ and $Q$ are open in the subspace $C$. They are disjoint, since $\operatorname{Int} A \subseteq A \subseteq \overline{A}$.

$P \cup Q = C$. Let $c \in C$. If $c \in \overline{A}$, then $c \notin \operatorname{Bd} A$ forces $c \notin \overline{X \setminus A}$, i.e. $c \in \operatorname{Int} A$, so $c \in P$. Otherwise $c \in X \setminus \overline{A}$, so $c \in Q$.

$P \neq \emptyset$. Pick $a \in C \cap A$. Then $a \in \overline{A}$, and as above $a \notin \operatorname{Bd} A$ gives $a \in \operatorname{Int} A$; hence $a \in P$.

$Q \neq \emptyset$. Pick $b \in C \cap (X \setminus A)$. Then $b \in \overline{X \setminus A}$, and $b \notin \operatorname{Bd} A$ forces $b \notin \overline{A}$; hence $b \in Q$.

Thus $P, Q$ form a separation of $C$, contradicting the connectedness of $C$. Therefore $C \cap \operatorname{Bd} A \neq \emptyset$. □

## Problem 4.10

Let $X$ and $Y$ be two connected subspaces and $A \subsetneq X$ and $B \subsetneq Y$. Prove that $(X \times Y) \setminus (A \times B)$ is connected.

### Solution

*submitted by* **Harisankar B**

Let $Z = (X \times Y) \setminus (A \times B)$. We use two standard facts:

(a) For fixed $x \in X$ the map $y \mapsto (x, y)$ is an embedding of $Y$ onto $\{x\} \times Y$ (Problem 4.1); hence $\{x\} \times Y$ is connected. Likewise $X \times \{y\}$ is connected for every $y \in Y$.

(b) A union of connected subspaces having a point in common is connected; in particular, the union of two connected subspaces that intersect is connected.

*Proof.* Since $A \subsetneq X$ and $B \subsetneq Y$, we can fix $x_0 \in X \setminus A$ and $y_0 \in Y \setminus B$. Note that a point $(x, y)$ lies in $A \times B$ only if $x \in A$ *and* $y \in B$; hence

$$Z = \{(x, y) \in X \times Y : x \notin A \text{ or } y \notin B\}.$$

For $x \in X \setminus A$ and $y \in Y \setminus B$ define

$$T_x = (\{x\} \times Y) \cup (X \times \{y_0\}), \qquad S_y = (X \times \{y\}) \cup (\{x_0\} \times Y).$$

*Each $T_x$ and $S_y$ lies in $Z$.* Points of $\{x\} \times Y$ have first coordinate $x \notin A$; points of $X \times \{y_0\}$ have second coordinate $y_0 \notin B$. Similarly for $S_y$, using $y \notin B$ and $x_0 \notin A$.

*Each $T_x$ and $S_y$ is connected.* $T_x$ is the union of the connected sets $\{x\} \times Y$ and $X \times \{y_0\}$ (fact (a)), which meet at $(x, y_0)$; so $T_x$ is connected by fact (b). Likewise $S_y$ is the union of the connected sets $X \times \{y\}$ and $\{x_0\} \times Y$, which meet at $(x_0, y)$.

*They share a common point.* Every $T_x$ contains $X \times \{y_0\} \ni (x_0, y_0)$, and every $S_y$ contains $\{x_0\} \times Y \ni (x_0, y_0)$.

*They cover $Z$.* If $(x, y) \in Z$ with $x \notin A$, then $(x, y) \in \{x\} \times Y \subseteq T_x$; if $y \notin B$, then $(x, y) \in X \times \{y\} \subseteq S_y$.

Therefore

$$Z = \bigcup_{x \in X \setminus A} T_x \cup \bigcup_{y \in Y \setminus B} S_y$$

is a union of connected subspaces all containing the point $(x_0, y_0)$, and hence $Z$ is connected by fact (b). □

## Problem 4.11

Let $Y \subset X$; and let $X$ and $Y$ be connected. Prove that if $A$ and $B$ form a separation of $X \setminus Y$, then both $Y \cup A$ and $Y \cup B$ are connected.

### Solution

*submitted by* **Harisankar B**

We use the following characterisation of separations (Munkres, Lemma 23.1): *if $W$ is a subspace of $X$, then a separation of $W$ is a pair of disjoint non-empty sets whose union is $W$, neither of which contains a limit point (in $X$) of the other.* We also use that if $Y$ is connected and $W = C \cup D$ is a separation with $Y \subseteq W$, then $Y$ lies entirely in $C$ or entirely in $D$.

*Proof.* We show that $Y \cup A$ is connected; the argument for $Y \cup B$ is identical with $A$ and $B$ interchanged.

Suppose, for contradiction, that $C, D$ form a separation of $Y \cup A$. Since $Y$ is connected, $Y$ lies entirely in one of them; say $Y \subseteq C$. Then $D \subseteq (Y \cup A) \setminus Y \subseteq A$. We claim that

$$X = (C \cup B) \cup D$$

is a separation of $X$.

*Union and disjointness.* $C \cup D = Y \cup A$ and $A \cup B = X \setminus Y$, so $(C \cup B) \cup D = Y \cup A \cup B = X$. Moreover $D \cap C = \emptyset$, and $D \cap B \subseteq A \cap B = \emptyset$.

*Non-emptiness.* $D \neq \emptyset$ and $C \neq \emptyset$, as $C, D$ is a separation; hence $C \cup B \neq \emptyset$.

*No limit points across.* By Lemma 23.1 applied to the separations $C, D$ of $Y \cup A$ and $A, B$ of $X \setminus Y$:

- $D$ contains no limit point of $C$, and $C$ contains no limit point of $D$;
- $A$ contains no limit point of $B$, and $B$ contains no limit point of $A$.

Since $D \subseteq A$, the set $D$ contains no limit point of $B$; so $D$ contains no limit point of $C \cup B$ (a limit point of a union is a limit point of one of the sets). Also every limit point of $D$ is a limit point of $A$, so $B$ contains no limit point of $D$; together with the first bullet, $C \cup B$ contains no limit point of $D$.

By Lemma 23.1, $C \cup B$ and $D$ form a separation of $X$, contradicting the connectedness of $X$. Hence $Y \cup A$ is connected, and similarly $Y \cup B$ is connected. □

# Problem Set 5

## Problem 5.1

Prove that the punctured Euclidean space $X = \mathbb{R}^n \setminus \{0\}$ is path connected if and only if $n > 1$.

### Solution

*submitted by* **Mayuri**

We show that $X = \mathbb{R}^n \setminus \{0\}$ is path connected if and only if $n > 1$ (here $n \geq 1$).

*Case* $n = 1$. Then $X = (-\infty, 0) \cup (0, \infty)$, a union of two disjoint nonempty open sets, so $X$ is not connected, hence not path connected. Directly: if $\gamma\colon [0,1] \to X$ were a path from $-1$ to $1$, the intermediate value theorem would give $t$ with $\gamma(t) = 0 \notin X$, a contradiction.

*Case* $n \geq 2$. Let $x, y \in X$. Recall the segment $[x, y] = \{(1-t)x + ty : t \in [0,1]\}$ is the image of a path (a straight line path).

If $0 \notin [x, y]$, the straight line path from $x$ to $y$ lies in $X$.

Otherwise $0 = (1-t)x + ty$ for some $t \in [0,1]$. Since $x, y \neq 0$ we have $t \in (0,1)$, so $x$ and $y$ are linearly dependent: $y = -cx$ with $c = \frac{1-t}{t} > 0$. Since $n \geq 2$, choose $z \in \mathbb{R}^n$ linearly independent from $x$ (then also from $y$, as $y$ is a nonzero multiple of $x$). In particular $z \neq 0$.

**Claim.** $0 \notin [x, z]$.

If $0 = (1-s)x + sz$ with $s \in [0,1]$, then $s \neq 0$ (as $x \neq 0$) and $s \neq 1$ (as $z \neq 0$), so $z = -(\frac{1-s}{s})x$, contradicting linear independence. The same argument shows $0 \notin [z, y]$.

Hence the segment from $x$ to $z$ followed by the segment from $z$ to $y$ is a path in $X$ from $x$ to $y$. So $X$ is path connected. □

## Problem 5.2

Prove that $\mathbb{R}^2 \setminus \mathbb{Q}^2$ with the subspace topology of $\mathbb{R}^2$ is connected, but $\mathbb{R} \setminus \mathbb{Q}$ with the subspace topology of $\mathbb{R}$ is not connected.

### Solution

*submitted by* **Mayuri**

$\mathbb{R}^2 \setminus \mathbb{Q}^2$ *is connected.* We show it is path connected, which implies connected. Let $x, y \in \mathbb{R}^2 \setminus \mathbb{Q}^2$ with $x \neq y$.

For a point $p \in \mathbb{R}^2$, consider the lines through $p$; they are parametrized by their direction angle $\theta \in [0, \pi)$, so there are uncountably many. For each $q \in \mathbb{Q}^2$ there is exactly one line through $x$ and $q$ (note $q \neq x$ since $x \notin \mathbb{Q}^2$). Since $\mathbb{Q}^2$ is countable, only countably many lines through $x$ meet $\mathbb{Q}^2$. So all but countably many lines through $x$ are contained entirely in $\mathbb{R}^2 \setminus \mathbb{Q}^2$. The same holds for $y$.

Choose a line $L_x$ through $x$ missing $\mathbb{Q}^2$. Then choose a line $L_y$ through $y$ missing $\mathbb{Q}^2$ with a direction different from that of $L_x$ (possible, since uncountably many directions are available and only countably many are bad, and we exclude just one). Then $L_x$ and $L_y$ are not parallel, so they meet at a single point $p$. Go from $x$ to $p$ along $L_x$, then from $p$ to $y$ along $L_y$. This is a path in $\mathbb{R}^2 \setminus \mathbb{Q}^2$ from $x$ to $y$. Hence $\mathbb{R}^2 \setminus \mathbb{Q}^2$ is path connected, and therefore connected.

$\mathbb{R}\setminus\mathbb{Q}$ *is not connected.* Let $U=(-\infty,0)\cap(\mathbb{R}\setminus\mathbb{Q})$ and $V=(0,\infty)\cap(\mathbb{R}\setminus\mathbb{Q})$. Both are open in the subspace $\mathbb{R}\setminus\mathbb{Q}$ and they are disjoint. Since $0\in\mathbb{Q}$, we have $0\notin\mathbb{R}\setminus\mathbb{Q}$, so $U\cup V=\mathbb{R}\setminus\mathbb{Q}$. Both are nonempty ($-\sqrt{2}\in U$, $\sqrt{2}\in V$). So $\mathbb{R}\setminus\mathbb{Q}$ is the union of two disjoint nonempty open sets, hence disconnected. □

## Problem 5.3

Let $X$ and $Y$ be topological spaces, and suppose there exist embeddings $f\colon X\to Y$ and $g\colon Y\to X$. Is it necessary that $X$ and $Y$ are homeomorphic? Prove or provide a counterexample.

### Solution

*submitted by* **Mayuri**

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No, it is not necessary.

*Counterexample.* Let $X=[0,1]$ and $Y=(0,1)$, both with the subspace topology from $\mathbb{R}$.

- $f\colon X\to Y$, $f(x)=\frac{1}{4}+\frac{x}{2}$. This is continuous and injective, with image $[\frac{1}{4},\frac{3}{4}]$, and the inverse $[\frac{1}{4},\frac{3}{4}]\to[0,1]$, $y\mapsto 2y-\frac{1}{2}$, is continuous. So $f$ is a homeomorphism onto its image, i.e. an embedding.
- $g\colon Y\to X$, the inclusion $g(y)=y$. It is an embedding, since the topology on $Y$ is the subspace topology inherited from $\mathbb{R}$, which is also the subspace topology inherited from $X$.

But $X$ and $Y$ are not homeomorphic: $X=[0,1]$ is compact (Heine–Borel), whereas $Y=(0,1)$ is not, since the open cover $\{(\frac{1}{n},1):n\geq 2\}$ has no finite subcover (a finite subfamily has union $(\frac{1}{N},1)$ for the largest $n=N$ used, which misses $\frac{1}{2N}$, a point of $Y$). Compactness is preserved by homeomorphisms, so $X\ncong Y$. □

## Problem 5.4

Let $X$ be a non-empty topological space and let $\mathscr{B}$ be a base for the topology on $X$. Prove that $X$ is compact if and only if every open cover of $X$ consisting of elements of $\mathscr{B}$ has a finite subcover.

### Solution

*submitted by* **Mayuri**

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Let $\mathscr{B}$ be a base for the topology on $X$.

**($\Longrightarrow$)** If $X$ is compact, then every open cover of $X$ has a finite subcover. A cover consisting of elements of $\mathscr{B}$ is in particular an open cover (basis elements are open), so it has a finite subcover.

**($\Longleftarrow$)** Suppose every open cover of $X$ consisting of elements of $\mathscr{B}$ has a finite subcover. Let $\mathcal{U}$ be an arbitrary open cover of $X$. For each $x\in X$ choose $U_x\in\mathcal{U}$ with $x\in U_x$, and since $U_x$ is open and $\mathscr{B}$ is a base, choose $B_x\in\mathscr{B}$ with $x\in B_x\subseteq U_x$.

Then $\{B_x : x\in X\}$ is a cover of $X$ (each $x\in B_x$) by elements of $\mathscr{B}$. By hypothesis, there are $x_1,\dots,x_n\in X$ with $X=B_{x_1}\cup\cdots\cup B_{x_n}$. Since $B_{x_i}\subseteq U_{x_i}$, we get

$$X=U_{x_1}\cup\cdots\cup U_{x_n},$$

so $\{U_{x_1},\dots,U_{x_n}\}$ is a finite subcover of $\mathcal{U}$. Hence $X$ is compact. □

## Problem 5.5

Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on $X$ such that both $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are compact Hausdorff spaces. Prove that either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable.

### Solution

*submitted by* **Prabhu**

*Proof.* Suppose $\mathcal{T}_1$ and $\mathcal{T}_2$ are comparable. Without loss of generality, let

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Consider the identity map

$$I \colon (X, \mathcal{T}_2) \to (X, \mathcal{T}_1), \qquad I(x) := x.$$

Clearly, $I$ is *bijective*.

For all $U \in \mathcal{T}_1$, we have

$$I^{-1}(U) = U \in \mathcal{T}_2.$$

Hence, $I$ is *continuous*.

Since $I$ is a bijective continuous map from a compact space to a Hausdorff space, $I$ is a *homeomorphism*. So, $I^{-1}$ is also continuous.

Thus, for all $U \in \mathcal{T}_2$, we have

$$(I^{-1})^{-1}(U) = I(U) = U \in \mathcal{T}_1.$$

Hence,

$$\mathcal{T}_2 \subseteq \mathcal{T}_1.$$

Therefore,

$$\boxed{\mathcal{T}_1 = \mathcal{T}_2.}$$

Hence, either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable. □

## Problem 5.6

Prove that every compact subspace of a metric space is closed and bounded in that metric topology. Find an example of a metric space in which a closed and bounded subspace is not compact.

### Solution

*submitted by* **Prabhu**

*Proof.* Let $(X, d)$ be a metric space and let $K \subseteq X$ be compact.

We know that every metric space is Hausdorff and a compact subspace of a Hausdorff space is closed. Since $X$ is a metric space it is Hausdorff. And $K$ is a compact subspace of $X$, thus it is *closed* in $X$.

Let $a \in X$. Consider the collection

$$\mathcal{U} := \{B(a, n) : n \in \mathbb{N}\},$$

where

$$B(a,n) := \{x \in X : d(x,a) < n\}.$$

For all $x \in K$, $d(x,a)$ is finite; so, $\exists\, n \in \mathbb{N}$ such that $d(x,a) < n$; i.e., $x \in B(a,n)$. Thus, $\mathcal{U}$ is an open cover of $A$.

Since $K$ is compact, there exists a finite subcover $\{B(a,n_1),\ldots,B(a,n_m)\}$. Therefore,

$$K \subseteq \bigcup_{i=1}^{m} B(a,n_i).$$

Let $N := \max\{n_1,n_2,\ldots,n_m\}$. Since $B(a,n_i) \subseteq B(a,N)$ for every $i$, we get

$$K \subseteq \bigcup_{i=1}^{m} B(a,n_i) \subseteq B(a,N).$$

Therefore, $K$ is *bounded*. Hence,

$$\boxed{K \text{ is closed and bounded.}}$$

□

**Example:** Consider the metric space $X := (0,1)$ with the usual metric ($\because d(x,y) := |x-y|$), and let $A := X$. Clearly, $A$ is closed ($\because$ it is the entire space) and bounded ($\because \forall x \in A,\ 0 < x < 1$) in $X$.

However, consider the collection

$$\mathcal{U} = \left\{ \left(\frac{1}{n}, 1\right) : n \in \mathbb{N},\ n \geq 2 \right\}.$$

Each member of $\mathcal{U}$ is open in the subspace $(0,1)$. And,

$$(0,1) = \bigcup_{n=2}^{\infty} \left(\frac{1}{n}, 1\right).$$

Thus, $\mathcal{U}$ is an open cover of $A$.

For every finite subcollection $\left\{\left(\frac{1}{n_1},1\right), \left(\frac{1}{n_2},1\right), \ldots, \left(\frac{1}{n_k},1\right)\right\}$, let $N = \max\{n_1,n_2,\ldots,n_k\}$. Since $n_i \leq N$ for every $i = 1,\ldots,k$, we have

$$\left(\frac{1}{n_i},1\right) \subseteq \left(\frac{1}{N},1\right).$$

Therefore,

$$\bigcup_{i=1}^{k} \left(\frac{1}{n_i},1\right) \subseteq \left(\frac{1}{N},1\right) \subsetneq (0,1).$$

It cannot cover $(0,1)$, since points in $(0,1/N]$ remain uncovered. Hence, $\mathcal{U}$ has no finite subcover. Therefore,

$$\boxed{(0,1) \text{ is closed and bounded but not compact in } (0,1).}$$

### Problem 5.7

Let $X$ be a compact metric space. Prove that if $f\colon X \to X$ is an isometry (distance preserving map) of $X$, then $f$ is a homeomorphism.

**Solution pending**

### Problem 5.8

Let $\mathbb{R}$ be given the cocountable topology. Is the closed interval $[0, 1]$ a compact subspace of $\mathbb{R}$ under this topology?

**Solution pending**

### Problem 5.9

Let $f\colon X \to Y$ be a continuous map between topological spaces. Show that if $X$ is compact and $Y$ is Hausdorff, then $f$ is a closed map.

**Solution pending**

### Problem 5.10

Prove that if $Y$ is compact, then the projection $\pi_X\colon X \times Y \to X$ is a closed map.

**Solution pending**

# Problem Set 6

## Problem 6.1

Let $p\colon X \twoheadrightarrow Y$ be a closed continuous surjective map such that $p^{-1}(y)$ is compact for each $y \in Y$. Prove that if $Y$ is compact, then so is $X$.

**Solution** *submitted by* **Yashpal Prasad**

*Proof.* Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an arbitrary open covering of the space $X$. We wish to extract a finite subcovering of $X$ from $\mathcal{U}$ to establish that $X$ is compact. For each point $y \in Y$, the fiber $p^{-1}(y) \subseteq X$ is a compact subspace by hypothesis. Since $\mathcal{U}$ covers $X$, it covers $p^{-1}(y)$ in particular; hence, we can extract a finite subcollection of open sets from $\mathcal{U}$, say $U_{\alpha_{1,y}}, U_{\alpha_{2,y}}, \ldots, U_{\alpha_{k_y,y}}$, which covers $p^{-1}(y)$.

Define the union of these selected open sets as $S_y = \bigcup_{i=1}^{k_y} U_{\alpha_{i,y}}$, so that $p^{-1}(y) \subseteq S_y$. The complement $V_y = X \setminus S_y$ is a closed subset of $X$ because $S_y$ is a finite union of open sets. Because $p\colon X \to Y$ is a closed map, the image $p(V_y)$ is a closed subset of $Y$, which implies that its complement $W_y = Y \setminus p(V_y)$ is an open subset of $Y$.

We claim that $y \in W_y$. Suppose to the contrary that $y \notin W_y$. Then $y \in p(V_y)$, meaning there exists some $x \in V_y$ such that $p(x) = y$, or $x \in p^{-1}(y)$. But $V_y = X \setminus S_y$ is disjoint from $S_y$, whereas $p^{-1}(y) \subseteq S_y$, yielding a contradiction. Thus, $y \in W_y$ for every $y \in Y$.

The collection of open sets $\{W_y\}_{y \in Y}$ forms an open covering of $Y$. Since $Y$ is compact, this covering contains a finite subcovering, say $W_{y_1}, W_{y_2}, \ldots, W_{y_m}$. For each $j$, we have $p^{-1}(W_{y_j}) \subseteq X \setminus V_{y_j} = S_{y_j} = \bigcup_{i=1}^{k_{y_j}} U_{\alpha_{i,y_j}}$. Taking the inverse images over the finite subcovering yields:

$$X = p^{-1}(Y) = \bigcup_{j=1}^{m} p^{-1}(W_{y_j}) \subseteq \bigcup_{j=1}^{m} \left( \bigcup_{i=1}^{k_{y_j}} U_{\alpha_{i,y_j}} \right).$$

**Conclusion:** We have expressed $X$ as a finite union of sets belonging to $\mathcal{U}$. Therefore, $X$ is compact. □

## Problem 6.2

Let $X$ be a compact metric space. Prove that if $f\colon X \to X$ is an isometry (distance preserving map) of $X$, then $f$ is a homeomorphism.

### Solution

*submitted by* **Yashpal Prasad**

*Proof.* Let $f\colon X \to X$ satisfy $d(f(x), f(y)) = d(x, y)$ for all $x, y \in X$. A homeomorphism is a continuous bijection with continuous inverse. We check each part in turn.

**Step 1: $f$ is continuous.** Given $\epsilon > 0$, take $\delta = \epsilon$. If $d(x, y) < \delta$, then $d(f(x), f(y)) = d(x, y) < \epsilon$. So $f$ is (uniformly) continuous.

**Step 2: $f$ is injective.** If $f(x) = f(y)$, then $d(x, y) = d(f(x), f(y)) = 0$. Hence $x = y$.

**Step 3: $f$ is surjective (this is where compactness is used).** Suppose, for contradiction, that $f(X) \neq X$, and pick $y \in X \setminus f(X)$.

(a) *$f(X)$ is closed.* By Step 1, $f$ is continuous, and $X$ is compact. The continuous image of a compact set is compact (standard theorem), so $f(X)$ is compact. A compact subset of a metric space is closed, so $f(X)$ is closed.

(b) *$y$ is at positive distance from $f(X)$.* Since $y \notin f(X)$ and $f(X)$ is closed, some open ball $B(y, \epsilon)$ misses $f(X)$. That is, $d(y, z) \geq \epsilon$ for every $z \in f(X)$.

(c) *Build a sequence with no convergent subsequence.* Define $x_n = f^n(y)$ for $n \geq 0$, where $f^0 = \mathrm{id}$. Take $m > n \geq 0$. Since $f$ is an isometry, so is $f^n$, and therefore

$$d(x_n, x_m) = d\big(f^n(y), f^n(f^{m-n}(y))\big) = d\big(y, f^{m-n}(y)\big).$$

Because $m - n \geq 1$, the point $f^{m-n}(y) = f(f^{m-n-1}(y))$ lies in $f(X)$. By (b), $d(x_n, x_m) \geq \epsilon$.

(d) *Contradiction.* Every pair of distinct terms of $(x_n)$ is at distance at least $\epsilon$. So no subsequence is Cauchy, and hence none converges. But $X$ is a compact metric space, so it is sequentially compact (standard theorem), and $(x_n)$ must have a convergent subsequence. This is a contradiction.

Therefore $f(X) = X$, so $f$ is surjective.

**Step 4: $f^{-1}$ is continuous.** By Steps 2 and 3, $f$ is a bijection, so $f^{-1}\colon X \to X$ exists. For $a, b \in X$, write $a = f(x)$ and $b = f(y)$. Then

$$d\big(f^{-1}(a), f^{-1}(b)\big) = d(x, y) = d(f(x), f(y)) = d(a, b).$$

So $f^{-1}$ is also an isometry, and by the same argument as in Step 1 it is continuous.

**Conclusion:** $f$ is a continuous bijection with continuous inverse, so $f$ is a homeomorphism. □

## Problem 6.3

Prove that the composition of two quotient maps is a quotient map.

**Solution** *submitted by* **Yashpal Prasad**

*Proof.* Let $p\colon X \to Y$ and $q\colon Y \to Z$ be quotient maps. By definition, both $p$ and $q$ are continuous and surjective maps. Consider the composite map $qp\colon X \to Z$ defined by $(qp)(x) = q(p(x))$. The composition of continuous functions is continuous, and the composition of surjective functions is surjective; hence $qp$ is continuous and surjective.

To prove that $qp$ is a quotient map, we must show that a subset $U \subseteq Z$ is open in $Z$ if and only if its inverse image $(qp)^{-1}(U)$ is open in $X$. By set-theoretic properties of inverse images under compositions, $(qp)^{-1}(U) = p^{-1}(q^{-1}(U))$.

Suppose $U \subseteq Z$ is such that $(qp)^{-1}(U)$ is open in $X$. Since $p$ is a quotient map, a subset $V \subseteq Y$ is open in $Y$ if and only if $p^{-1}(V)$ is open in $X$. Setting $V = q^{-1}(U)$, we have $p^{-1}(V) = (qp)^{-1}(U)$ open in $X$, which implies $q^{-1}(U)$ is open in $Y$. Now, since $q$ is a quotient map and $q^{-1}(U)$ is open in $Y$, it follows that $U$ is open in $Z$. Conversely, if $U$ is open in $Z$, then $q^{-1}(U)$ is open in $Y$ by continuity of $q$, and $p^{-1}(q^{-1}(U))$ is open in $X$ by continuity of $p$.

**Conclusion:** $U$ is open in $Z$ if and only if $(qp)^{-1}(U)$ is open in $X$. Thus, $qp$ is a quotient map. □

## Problem 6.4

Prove that a surjective continuous map $f\colon X \twoheadrightarrow Y$ that is either an open map or a closed map is a quotient map.

**Solution** *submitted by* **Indrajeet Kumawat**

*Proof.* A surjective continuous map $f\colon X \to Y$ is a quotient map if for any subset $U \subseteq Y$, $U$ is open in $Y$ if and only if $f^{-1}(U)$ is open in $X$ (equivalently with closed sets). We analyze the two cases separately:

**Case 1 ($f$ is an open map):** Let $U \subseteq Y$ be any subset. If $U$ is open in $Y$, then $f^{-1}(U)$ is open in $X$ by continuity. Conversely, suppose $f^{-1}(U)$ is open in $X$. Since $f$ is an open map, the image $f(f^{-1}(U))$ is open in $Y$. Because $f$ is surjective, $f(f^{-1}(U)) = U$, so $U$ is open in $Y$.

**Case 2 ($f$ is a closed map):** Let $C \subseteq Y$ be any subset. If $C$ is closed in $Y$, then $f^{-1}(C)$ is closed in $X$ by continuity. Conversely, suppose $f^{-1}(C)$ is closed in $X$. Since $f$ is a closed map, $f(f^{-1}(C))$ is closed in $Y$. Since $f$ is surjective, $f(f^{-1}(C)) = C$, so $C$ is closed in $Y$.

**Conclusion:** In both cases, the quotient map condition holds. Therefore, $f$ is a quotient map. □

## Problem 6.5

Prove that a bijective quotient map is a homeomorphism.

**Solution** *submitted by* **Indrajeet Kumawat**

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*Proof.* Let $f\colon X \to Y$ be a quotient map that is also bijective. Because $f$ is a quotient map, it is continuous and surjective. Because $f$ is bijective, it possesses an inverse function $f^{-1}\colon Y \to X$. To prove that $f$ is a homeomorphism, we must show that $f^{-1}$ is continuous (since $f$ is already continuous).

Showing $f^{-1}$ is continuous is equivalent to showing that for any open set $U \subseteq X$, its inverse image under $f^{-1}$, which is $(f^{-1})^{-1}(U) = f(U)$, is open in $Y$. Let $U$ be an arbitrary open subset of $X$. Because $f$ is injective, $f^{-1}(f(U)) = U$. By definition, $f$ is a quotient map, meaning a subset $V \subseteq Y$ is open in $Y$ if and only if $f^{-1}(V)$ is open in $X$. Choosing $V = f(U)$, we have $f^{-1}(V) = U$, which is given to be open in $X$. Therefore, $V = f(U)$ is open in $Y$.

**Conclusion:** Since $f(U)$ is open in $Y$ for every open set $U \subseteq X$, the inverse map $f^{-1}$ is continuous. Thus, $f$ is a homeomorphism. □

## Problem 6.6

Let $f_1\colon X_1 \to Y_1$ and $f_2\colon X_2 \to Y_2$ be maps between topological spaces, and let $f_1 \times f_2\colon X_1 \times X_2 \to Y_1 \times Y_2$ denote the product map $(x_1, x_2) \mapsto (f_1(x_1), f_2(x_2))$.

(a) Prove that if $f_1$ and $f_2$ are continuous, then $f_1 \times f_2$ is continuous with respect to the product topologies.

(b) Give a counterexample to show that if $p_1\colon X_1 \to Y_1$ and $p_2\colon X_2 \to Y_2$ are quotient maps, $p_1 \times p_2$ need not be a quotient map.

**Solution** *submitted by* **Indrajeet Kumawat**

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**(a)** A sub-basis for the product topology on $Y_1 \times Y_2$ consists of sets of the form $U_1 \times U_2$, where $U_1 \subseteq Y_1$ and $U_2 \subseteq Y_2$ are open. To prove $f_1 \times f_2$ is continuous, we show that the inverse image of any basic open set $U_1 \times U_2$ is open in $X_1 \times X_2$. Let $(x_1, x_2) \in (f_1 \times f_2)^{-1}(U_1 \times U_2)$. Then $(f_1(x_1), f_2(x_2)) \in U_1 \times U_2$, which implies $f_1(x_1) \in U_1$ and $f_2(x_2) \in U_2$, or equivalently $x_1 \in f_1^{-1}(U_1)$ and $x_2 \in f_2^{-1}(U_2)$. Thus, $(f_1 \times f_2)^{-1}(U_1 \times U_2) = f_1^{-1}(U_1) \times f_2^{-1}(U_2)$. Since $f_1$ and $f_2$ are continuous, $f_1^{-1}(U_1)$ and $f_2^{-1}(U_2)$ are open in $X_1$ and $X_2$ respectively, making their Cartesian product open in $X_1 \times X_2$.

**Conclusion (a):** $f_1 \times f_2$ is continuous. □

**(b)** Consider the quotient map $p_1 = \pi\colon \mathbb{R} \to \mathbb{R}/\mathbb{N}$ obtained by collapsing the closed set of integers $\mathbb{N} = \{1, 2, 3, \dots\}$ to a single point $*$, and let $p_2 = \mathrm{id}_{\mathbb{Q}}\colon \mathbb{Q} \to \mathbb{Q}$ be the identity map on the rational numbers, which is a homeomorphism and hence a quotient map. We examine their product map $F = \pi \times \mathrm{id}_{\mathbb{Q}}\colon \mathbb{R} \times \mathbb{Q} \to (\mathbb{R}/\mathbb{N}) \times \mathbb{Q}$. By part (a), $F$ is continuous, and it is clearly surjective. However, we claim that $F$ is not a quotient map.

To see why, note that the fibers of $F$ over points in the codomain are either single points or sets of the form $\mathbb{N} \times \{s\}$ for $s \in \mathbb{Q}$. A subset $U \subseteq \mathbb{R} \times \mathbb{Q}$ is saturated with respect to $F$ if for each rational $s$, either the entire vertical slice $\mathbb{N} \times \{s\}$ is contained in $U$ or none of it is. We can construct a saturated open set $U$ by taking an enumeration of irrational numbers $r_n = \sqrt{2}/n$ for $n \geq 1$ and defining continuous functions $h_n(s) = \min\{1/4, |s - r_n|\}$ for $s \in \mathbb{Q}$. Because each $r_n$ is irrational, $h_n(s) > 0$ everywhere on $\mathbb{Q}$. We define the open set

$$U = \bigcup_{n=1}^{\infty} \{(x, s) \in \mathbb{R} \times \mathbb{Q} : |x - n| < h_n(s)\}.$$

Since $(n, s) \in U$ for every integer $n$ and rational $s$, the set contains $\mathbb{N} \times \mathbb{Q}$, ensuring it is saturated.

Let $W = F(U)$ be the image of $U$. Because $U$ is saturated, the inverse image $F^{-1}(W) = U$ is open in $\mathbb{R} \times \mathbb{Q}$. If $F$ were a quotient map, its image $W$ would necessarily be open in $(\mathbb{R}/\mathbb{N}) \times \mathbb{Q}$. Notice that the point $(*, 0)$ belongs to $W$. If $W$ were open, there would exist an open neighborhood $V$ of $*$ in $\mathbb{R}/\mathbb{N}$ and a positive number $\delta > 0$ such that $V \times ((-\delta, \delta) \cap \mathbb{Q}) \subseteq W$. The inverse image $\pi^{-1}(V)$ is an open neighborhood in $\mathbb{R}$ containing all integers $n \geq 1$, meaning for each $n$ there exists some $0 < \eta_n < 1/4$ such that $(n - \eta_n, n + \eta_n) \subseteq \pi^{-1}(V)$. By choosing $n$ sufficiently large such that $|r_n| < \delta/2$ and picking a rational $s$ close to $r_n$ via the density of $\mathbb{Q}$, we can construct a point $x = n + (|s - r_n| + \eta_n)/2$ that satisfies $|s - r_n| < |x - n| < \eta_n$ while simultaneously contradicting the bound established by $h_n(s)$. This contradiction shows that the image $W$ cannot be open in the product topology.

**Conclusion (b):** The product of quotient maps $p_1 \times p_2$ is not necessarily a quotient map. □

## Problem 6.7

Let $X$ and $Y$ be topological spaces, and $f\colon X \to Y$ be a bijective continuous map. Show that if $Y$ is Hausdorff, then $X$ is also Hausdorff.

### Solution

*submitted by* **Himanshu Sharma**

*Proof.* To show that $X$ is Hausdorff, we must demonstrate that for any two distinct points $x_1, x_2 \in X$ with $x_1 \neq x_2$, there exist disjoint open neighborhoods $U_1, U_2 \subseteq X$ containing $x_1$ and $x_2$ respectively. Since $f\colon X \to Y$ is injective, $f(x_1) \neq f(x_2)$. Because $Y$ is a Hausdorff space, there exist disjoint open sets $V_1, V_2 \subseteq Y$ such that $f(x_1) \in V_1$, $f(x_2) \in V_2$, and $V_1 \cap V_2 = \emptyset$.

Since $f$ is continuous, the inverse images $U_1 = f^{-1}(V_1)$ and $U_2 = f^{-1}(V_2)$ are open subsets of $X$ containing $x_1$ and $x_2$ respectively. Furthermore, their intersection is:

$$U_1 \cap U_2 = f^{-1}(V_1) \cap f^{-1}(V_2) = f^{-1}(V_1 \cap V_2) = f^{-1}(\emptyset) = \emptyset.$$

**Conclusion:** We have found disjoint open neighborhoods separating $x_1$ and $x_2$. Therefore, $X$ is Hausdorff. □

## Problem 6.8

Let $X$ be a compact topological space and $p\colon X \twoheadrightarrow Y$ be a quotient map.

(a) Prove that $Y$ is compact.

(b) Show that if $Y$ is Hausdorff, then $p$ is a closed map.

**Solution** *submitted by* **Himanshu Sharma**

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**(a)** A quotient map $p\colon X \to Y$ is continuous and surjective. It is a fundamental topological theorem that the continuous image of a compact space is compact. Since $X$ is compact and $Y = p(X)$, $Y$ is compact.

**Conclusion (a):** $Y$ is compact. □

**(b)** Let $C \subseteq X$ be an arbitrary closed set in $X$. We must prove that $p(C)$ is closed in $Y$. Because $X$ is compact and $C$ is closed in $X$, $C$ is compact. Consider the restriction map $p|_C\colon C \to p(C)$, which is continuous. Thus, $p(C) = p|_C(C)$ is the continuous image of a compact space, so $p(C)$ is compact in $Y$. Since $Y$ is Hausdorff, any compact subspace of a Hausdorff space is closed. Consequently, $p(C)$ is closed in $Y$.

**Conclusion (b):** $p$ is a closed map. □

## Problem 6.9

Let $p\colon X \to Y$ be a continuous surjection. Prove that $p$ is a closed map if and only if for every subset $S \subseteq Y$ and every open set $U \subseteq X$ containing $p^{-1}(S)$, there exists an open set $V \subseteq Y$ containing $S$ such that $p^{-1}(V) \subseteq U$.

**Solution** *submitted by* **Himanshu Sharma**

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*Proof.* **Proof of (⇒):** Suppose $p$ is a closed map. Let $S \subseteq Y$ and let $U \subseteq X$ be an open set containing $p^{-1}(S)$. Consider the closed set $C = X \setminus U$. Since $p$ is a closed map, $p(C)$ is closed in $Y$, so $V = Y \setminus p(C)$ is open in $Y$. For any $s \in S$, $p^{-1}(\{s\}) \subseteq p^{-1}(S) \subseteq U$, meaning no point in the fiber belongs to $C$. Thus $s \notin p(C)$, so $s \in V$, implying $S \subseteq V$. Furthermore, $p^{-1}(V) = X \setminus p^{-1}(p(C)) \subseteq X \setminus C = U$.

**Proof of (⇐):** Conversely, suppose the condition holds. Let $C \subseteq X$ be closed. We must show $p(C)$ is closed in $Y$, or $Y \setminus p(C)$ is open. Let $y \in Y \setminus p(C)$ and set $S = \{y\}$. The fiber $p^{-1}(\{y\})$ is disjoint from $C$, so $p^{-1}(\{y\}) \subseteq X \setminus C$. Let $U = X \setminus C$, which is open. By hypothesis, there exists an open set $V_y \subseteq Y$ containing $y$ such that $p^{-1}(V_y) \subseteq U = X \setminus C$. This implies $V_y \cap p(C) = \emptyset$, so $V_y \subseteq Y \setminus p(C)$. Thus, $Y \setminus p(C)$ is open.

**Conclusion:** $p$ is a closed map if and only if the stated condition holds. □

# Problem Set 7 (Munkres §22)

## Problem 7.1

Check the details of Example 3.[a]

[a] Example 3 of §22 (The Quotient Topology) in J. R. Munkres, *Topology*, 2nd ed.

**Solution** *submitted by* **Karan Gangadharan**

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The quotient topology on $A$ induced by $p$ is given by

$$\tau_A = \{U \subseteq A : p^{-1}(U) \text{ is open in } \mathbb{R}\}.$$

Since

$$p^{-1}(\{a\}) = (-\infty, 0), \qquad p^{-1}(\{b\}) = (0, \infty), \qquad p^{-1}(\{c\}) = \{0\},$$

and

$$p^{-1}(\{a, b\}) = \mathbb{R} \setminus \{0\},$$

we observe that the open subsets of $A$ are

$$\varnothing, \quad \{a\}, \quad \{b\}, \quad \{a, b\}, \quad \{a, b, c\}.$$

Hence, the quotient topology on $A$ is

$$\boxed{\tau_A = \{\varnothing, \{a\}, \{b\}, \{a, b\}, \{a, b, c\}\}.}$$

## Problem 7.2

(a) Let $p\colon X \to Y$ be a continuous map. Show that if there is a continuous map $f\colon Y \to X$ such that $p \circ f$ equals the identity map of $Y$, then $p$ is a quotient map.

(b) If $A \subset X$, a *retraction* of $X$ onto $A$ is a continuous map $r\colon X \to A$ such that $r(a) = a$ for each $a \in A$. Show that a retraction is a quotient map.

**Solution** *submitted by* **Karan Gangadharan**

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*Proof.* **(a)** Let $p\colon X \to Y$ be a continuous map. Suppose there exists a continuous map $f\colon Y \to X$ such that

$$p \circ f = I_Y,$$

where $I_Y$ is the identity map on $Y$.

We know that $p$ is continuous. Also, since

$$p(f(y)) = y, \qquad \forall y \in Y,$$

the map $p$ is surjective.

Now, let $U \subseteq Y$ be such that $p^{-1}(U)$ is open in $X$. Since $f$ is continuous, we have $f^{-1}\big(p^{-1}(U)\big)$ open in $Y$. But

$$f^{-1}\big(p^{-1}(U)\big) = (p \circ f)^{-1}(U) = I_Y^{-1}(U) = U.$$

Therefore, $U$ is open in $Y$.

Thus, $p$ is a continuous surjective map satisfying

$$U \text{ is open in } Y \iff p^{-1}(U) \text{ is open in } X.$$

Hence, $p$ is a quotient map.

$\boxed{p \text{ is a quotient map.}}$

**(b)** Let $A \subseteq X$ and let $r\colon X \to A$ be a retraction such that

$$r(a) = a, \qquad \forall a \in A.$$

We know that $r$ is continuous by definition. Also, since $r(a) = a$ for every $a \in A$, the map $r$ is surjective.

Now, let $U \subseteq A$ be such that $r^{-1}(U)$ is open in $X$. Since $A$ has the subspace topology, we have $r^{-1}(U) \cap A$ open in $A$. But, using the property of retraction,

$$\begin{aligned} r^{-1}(U) \cap A &= \{x \in A : r(x) \in U\} \\ &= \{x \in A : x \in U\} \\ &= U. \end{aligned}$$

Therefore, $U$ is open in $A$.

Thus, $r$ is a continuous surjective map such that

$$U \text{ is open in } A \iff r^{-1}(U) \text{ is open in } X.$$

Hence, $r$ is a quotient map.

$\boxed{\text{Every retraction is a quotient map.}}$ □

## Problem 7.3

Let $\pi_1\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be projection on the first coordinate. Let $A$ be the subspace of $\mathbb{R} \times \mathbb{R}$ consisting of all points $x \times y$ for which either $x \geq 0$ or $y = 0$ (or both); let $q\colon A \to \mathbb{R}$ be obtained by restricting $\pi_1$. Show that $q$ is a quotient map that is neither open nor closed.

### Solution

*submitted by* **Karan Gangadharan**

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*Proof.* Let

$$\pi_1\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$

be the projection onto the first coordinate, defined by $\pi_1(x, y) = x$. Let

$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ or } y = 0\}.$$

Define $g\colon A \to \mathbb{R}$ by restricting $\pi_1$ to $A$. Thus, $g(x, y) = x$.

We shall show that $g$ is a quotient map but is neither open nor closed.

**(i) $g$ is a quotient map:** Since $g$ is the restriction of the continuous projection map $\pi_1$, it is continuous. Also, $g$ is surjective, because for every $x \in \mathbb{R}$,

$$(x, 0) \in A \quad \text{and} \quad g(x, 0) = x.$$

Now, consider the map $f\colon \mathbb{R} \to A$ defined by $f(x) = (x, 0)$. Clearly, $f$ is continuous and

$$(g \circ f)(x) = g(x, 0) = x.$$

Therefore, $g \circ f = I_{\mathbb{R}}$. By Problem 7.2(a), since $g$ and $f$ are continuous and $g \circ f = I_{\mathbb{R}}$, it follows that

$$\boxed{g \text{ is a quotient map.}}$$

**(ii) $g$ is not an open map:** Consider the set

$$U = A \cap ((-1, 1) \times (1/2, 3/2)).$$

Since $(-1, 1) \times (1/2, 3/2)$ is open in $\mathbb{R}^2$, $U$ is open in $A$. Now, by the definition of $A$, if $(x, y) \in U$, then $x \geq 0$, since $y \neq 0$. Consequently,

$$g(U) = [0, 1).$$

But $[0, 1)$ is not open in $\mathbb{R}$. Thus, $g$ does not map every open subset of $A$ onto an open subset of $\mathbb{R}$. Hence,

$$\boxed{g \text{ is not an open map.}}$$

**(iii) $g$ is not a closed map:** Consider the subset

$$F = \left\{ \left( \frac{1}{n}, n \right) : n \in \mathbb{N} \right\}.$$

Clearly, $F \subseteq A$, since $1/n > 0$ for every $n \in \mathbb{N}$. We claim that $F$ is closed in $A$. Indeed, $F$ is closed in $\mathbb{R}^2$, because the second coordinates of its points tend to infinity, so it has no accumulation point in $\mathbb{R}^2$. Therefore, $F$ is also closed in the subspace $A$. Now,

$$g(F) = g\left( \left\{ \left( \frac{1}{n}, n \right) : n \in \mathbb{N} \right\} \right) = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$

But

$$\frac{1}{n} \longrightarrow 0 \quad \text{as } n \longrightarrow \infty,$$

and $0 \notin g(F)$. Hence, $g(F)$ is not closed in $\mathbb{R}$. Therefore, $g$ does not map every closed subset of $A$ onto a closed subset of $\mathbb{R}$. Thus,

$$\boxed{g \text{ is not a closed map.}}$$

Combining (i), (ii) and (iii), we conclude that

$$\boxed{g\colon A \to \mathbb{R} \text{ is a quotient map which is neither open nor closed.}} \qquad \square$$

## Problem 7.4

(a) Define an equivalence relation on the plane $X = \mathbb{R}^2$ as follows:

$$x_0 \times y_0 \sim x_1 \times y_1 \quad \text{if} \quad x_0 + y_0^2 = x_1 + y_1^2.$$

Let $X^*$ be the corresponding quotient space. It is homeomorphic to a familiar space; what is it? [*Hint:* Set $g(x \times y) = x + y^2$.]

(b) Repeat (a) for the equivalence relation

$$x_0 \times y_0 \sim x_1 \times y_1 \quad \text{if} \quad x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

**Solution** *submitted by* **Karan Gangadharan**

*Proof.* **(a)** Define an equivalence relation on $X = \mathbb{R}^2$ by

$$(x_0, y_0) \sim (x_1, y_1) \iff x_0 + y_0^2 = x_1 + y_1^2.$$

Let $X^* = X/\sim$ be the corresponding quotient space. We shall show that $X^*$ is homeomorphic to $\mathbb{R}$.

Define a map $g\colon X \to \mathbb{R}$ by

$$g(x, y) = x + y^2.$$

Since $g$ is a polynomial function, it is continuous. Also, $g$ is surjective, since for every $t \in \mathbb{R}$, $g(t, 0) = t$.

Now, define a map $f\colon \mathbb{R} \to X$ by $f(t) = (t, 0)$. Clearly, $f$ is continuous and

$$(g \circ f)(t) = g(t, 0) = t.$$

Thus, $g \circ f = I_{\mathbb{R}}$. Therefore, by Problem 7.2(a), $g$ is a quotient map.

Let $q\colon X \to X^*$ be the natural quotient map defined by $q(x, y) = [(x, y)]$. Define $\overline{g}\colon X^* \to \mathbb{R}$ by

$$\overline{g}([(x, y)]) = x + y^2.$$

We show that $\overline{g}$ is a homeomorphism.

*(i) $\overline{g}$ is well-defined:* Suppose $[(x_0, y_0)] = [(x_1, y_1)]$. Then, by the definition of the equivalence relation, $x_0 + y_0^2 = x_1 + y_1^2$. Hence, $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Therefore, $\overline{g}$ is well-defined.

*(ii) $\overline{g}$ is bijective:* Suppose $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Then $x_0 + y_0^2 = x_1 + y_1^2$. Thus, $(x_0, y_0) \sim (x_1, y_1)$, which implies $[(x_0, y_0)] = [(x_1, y_1)]$. Therefore, $\overline{g}$ is injective. Further, since $g$ is surjective, for every $t \in \mathbb{R}$, there exists $(x, y) \in X$ such that $\overline{g}([(x, y)]) = t$. Hence, $\overline{g}$ is surjective. Therefore, $\overline{g}$ is bijective.

*(iii) $\overline{g}$ is a homeomorphism:* We have

$$g = \overline{g} \circ q.$$

Since $g$ is continuous and $q$ is a quotient map, the quotient property of $q$ implies that $\overline{g}$ is continuous.

Also, let $U$ be any open subset of $X^*$. Since $\overline{g}$ is injective, we have

$$g^{-1}(\overline{g}(U)) = q^{-1}(U).$$

Since $q$ is a quotient map and $U$ is open in $X^*$, $q^{-1}(U)$ is open in $X$. Thus, $g^{-1}(\overline{g}(U))$ is open in $X$. Since $g$ is a quotient map, $\overline{g}(U)$ is open in $\mathbb{R}$. Therefore, $\overline{g}$ is an open map.

Hence, $\overline{g}$ is a continuous bijective open map, and consequently it is a homeomorphism. Thus,

$$\boxed{X^* \cong \mathbb{R}.}$$

**(b)** Define an equivalence relation on $X = \mathbb{R}^2$ by

$$(x_0, y_0) \sim (x_1, y_1) \iff x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

Let $X^* = X/\sim$ be the corresponding quotient space. We shall show that $X^*$ is homeomorphic to $[0, \infty)$.

Define a map $g\colon X \to [0, \infty)$ by

$$g(x, y) = x^2 + y^2.$$

Since $g$ is a polynomial function, it is continuous. Also, $g$ is surjective, since for every $t \in [0, \infty)$, $g(\sqrt{t}, 0) = t$.

Now, define a map $f\colon [0, \infty) \to X$ by $f(t) = (\sqrt{t}, 0)$. Clearly, $f$ is continuous and

$$(g \circ f)(t) = g(\sqrt{t}, 0) = t.$$

Thus, $g \circ f = I_{[0,\infty)}$. Therefore, by Problem 7.2(a), $g$ is a quotient map.

Let $q\colon X \to X^*$ be the natural quotient map defined by $q(x, y) = [(x, y)]$. Define $\overline{g}\colon X^* \to [0, \infty)$ by

$$\overline{g}([(x, y)]) = x^2 + y^2.$$

We show that $\overline{g}$ is a homeomorphism.

*(i) $\overline{g}$ is well-defined:* Suppose $[(x_0, y_0)] = [(x_1, y_1)]$. Then, by the definition of the equivalence relation, $x_0^2 + y_0^2 = x_1^2 + y_1^2$. Hence, $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Therefore, $\overline{g}$ is well-defined.

*(ii) $\overline{g}$ is bijective:* Suppose $\overline{g}([(x_0, y_0)]) = \overline{g}([(x_1, y_1)])$. Then $x_0^2 + y_0^2 = x_1^2 + y_1^2$. Thus, $(x_0, y_0) \sim (x_1, y_1)$, which implies $[(x_0, y_0)] = [(x_1, y_1)]$. Therefore, $\overline{g}$ is injective. Further, since $g$ is surjective, for every $t \in [0, \infty)$, there exists $(x, y) \in X$ such that $\overline{g}([(x, y)]) = t$. Hence, $\overline{g}$ is surjective. Therefore, $\overline{g}$ is bijective.

*(iii) $\overline{g}$ is a homeomorphism:* We have

$$g = \overline{g} \circ q.$$

Since $g$ is continuous and $q$ is a quotient map, the quotient property of $q$ implies that $\overline{g}$ is continuous.

Also, let $U$ be any open subset of $X^*$. Since $\overline{g}$ is injective, we have

$$g^{-1}(\overline{g}(U)) = q^{-1}(U).$$

Since $q$ is a quotient map and $U$ is open in $X^*$, $q^{-1}(U)$ is open in $X$. Thus, $g^{-1}(\overline{g}(U))$ is open in $X$. Since $g$ is a quotient map, $\overline{g}(U)$ is open in $[0, \infty)$. Therefore, $\overline{g}$ is an open map.

Hence, $\overline{g}$ is a continuous bijective open map, and consequently it is a homeomorphism. Thus,

$$\boxed{X^* \cong [0, \infty).}$$ □

## Problem 7.5

Let $p\colon X \to Y$ be an open map. Show that if $A$ is open in $X$, then the map $q\colon A \to p(A)$ obtained by restricting $p$ is an open map.

**Solution pending**

## Problem 7.6

Recall that $\mathbb{R}_K$ denotes the real line in the $K$-topology (see §13). Let $Y$ be the quotient space obtained from $\mathbb{R}_K$ by collapsing the set $K$ to a point; let $p\colon \mathbb{R}_K \to Y$ be the quotient map.

(a) Show that $Y$ satisfies the $T_1$ axiom, but is not Hausdorff.

(b) Show that $p \times p\colon \mathbb{R}_K \times \mathbb{R}_K \to Y \times Y$ is not a quotient map. [*Hint:* The diagonal is not closed in $Y \times Y$, but its inverse image is closed in $\mathbb{R}_K \times \mathbb{R}_K$.]

**Solution pending**